Bacheloroppgave

LOG650 Forskningsprosjekt: Logistikk og KI

Long-Term Service Agreement (LTSA) versus Non-LTSA Maintenance for Marine Engines: A Case-Based Analysis of Maintenance Activity, Procurement-Related Cost Exposure, Availability Indicators, and Financial Risk in a Ship Management Portfolio

Forfatter: Evan Elias Håpnes

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Abstract

This thesis examines how an LTSA-based maintenance arrangement compares with a non-LTSA arrangement for marine engines within the Wilhelmsen Ship Management portfolio. The study compares two anonymized vessel cases: a Passenger Vessel operating under an LTSA structure and a Heavy Lift Vessel operating outside an LTSA structure. The purpose is to evaluate how the two maintenance arrangements differ in terms of engine-related maintenance activity, procurement activity, spare-parts support, operational availability indicators, and exposure to maintenance-related uncertainty.

The analysis is based only on engine-related BASSnet data for the period 2024–2025. The final data foundation includes job-history records, engine-related purchase-order records, spare-parts records, and defect records where available. Tableau financial dashboard data has been excluded from the final analysis because the available financial values were not sufficiently comparable between the two vessel cases and included broader vessel-level cost structures. The thesis therefore focuses on operational and procurement-related evidence from BASSnet rather than total vessel budgets or dashboard-based financial totals.

The corrected BASSnet dataset shows that the Passenger Vessel has 2,691 job-history records, 134 purchase-order rows, and 104 spare-parts records for the selected period. The Heavy Lift Vessel has 1,062 job-history records, 83 engine-related purchase-order rows, 121 spare-parts records, and 99 defect records. Because comparable defect data is not available for both vessels, defect records are used only as supporting context for the Heavy Lift Vessel and not as a direct defect-frequency comparison.

The results show that the two cases differ substantially in data structure, recorded maintenance activity, procurement activity, and spare-

parts support. However, these differences cannot be interpreted as direct evidence that one maintenance arrangement is universally better than the other. The Passenger Vessel and Heavy Lift Vessel differ in vessel type, operational profile, agreement structure, and data availability. The findings must therefore be interpreted as a case-based comparison of available BASSnet records rather than a complete financial or technical evaluation of all LTSA and non-LTSA strategies.

The thesis concludes that LTSA and non-LTSA maintenance arrangements should not be evaluated only by contract type. Instead, they should be assessed through a combined view of maintenance activity, procurement exposure, spare-parts support, operational availability indicators, and data quality. The findings are case-specific, but the structured comparison can support future maintenance-strategy evaluations under real operational data limitations.

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1.0 Introduction

The maritime industry depends on reliable technical systems to ensure safe, efficient, and predictable vessel operations. Marine engines are among the most critical technical assets onboard because their performance affects operational availability, maintenance planning, spare-parts demand, procurement activity, and maintenance-related cost exposure. Engine-related failures, delayed maintenance activities, or unavailable spare parts can therefore create both operational disruption and financial uncertainty. For this reason, marine engine maintenance is not only a technical issue, but also a strategic decision involving responsibility, support structure, cost exposure, and risk allocation.

Marine engine maintenance can be organized through different maintenance arrangements. One option is a Long-Term Service -based arrangement, where selected maintenance responsibilities, technical support, spare-parts support, and parts of the cost structure are governed through a long-term agreement with the engine manufacturer or service provider. Another option is a non-LTSA arrangement, where the operator or technical manager carries more direct responsibility for maintenance coordination, spare-parts planning, procurement, refurbishment decisions, and maintenance-related uncertainty. An LTSA-based arrangement may provide more predictable support and clearer risk allocation, while a non-LTSA arrangement may provide more flexibility and direct internal control.

This thesis compares these two maintenance arrangements within the Wilhelmsen Ship Management portfolio. The study uses two anonymized vessel cases: a Passenger Vessel operating under a LTSA and a Heavy Lift Vessel operating outside an LTSA structure. The two vessels are not treated as identical cases, as they differ in vessel type, operational profile, agreement structure, engine configuration, and data availability. The thesis is therefore designed as a case-based comparison of available engine-related BASSnet records, not as a direct comparison of identical vessels or as a universal evaluation of all LTSA and non-LTSA strategies.

In this thesis, BASSnet refers to the maintenance and procurement management system used as the source of the empirical data. The available BASSnet records include job-history data, purchase-order data, spare-parts data, and defect records where available. These records provide operational and procurement-related evidence about how engine maintenance activities were planned, registered, purchased, and followed up during the selected period. Since BASSnet is an operational system rather than a complete academic dataset, the records must be interpreted with attention to data availability, registration practice, and comparability between the two vessel cases.

The purpose of this thesis is to compare LTSA-based and non-LTSA maintenance arrangements using available engine-related BASSnet data from Wilhelmsen Ship Management. The comparison focuses on recorded maintenance activity, engine-related procurement activity, procurement-related cost exposure, spare-parts support, defect records where available, operational availability indicators, and financial risk exposure during the selected data period. The thesis does not attempt to calculate complete vessel-level cost or compare total vessel budgets. Instead, it uses available BASSnet indicators to support a structured maintenance strategy comparison.

1.1 Problem Statement

The main problem addressed in this thesis is formulated as follows:

How do LTSA-based and non-LTSA maintenance arrangements for MAN marine engines differ when compared through available BASSnet records on maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, and financial risk exposure?

This problem statement addresses a real maintenance strategy decision in asset-intensive maritime operations. It also reflects the actual empirical foundation of the thesis, which is limited to available engine-related BASSnet records for the 2024–2025 period.

Because the final dataset is based on BASSnet, cost exposure and financial risk are not analyzed as complete vessel-level totals. Instead, they are interpreted through available operational and procurement-related indicators, including job-history records, engine-related purchase-order records, spare-parts records, and defect records where available.

1.2 Sub Questions

To answer the main problem statement, the thesis is structured around the following sub-questions:

- How do the LTSA-based and non-LTSA cases differ in terms of recorded engine-related maintenance activity?
- How do the two cases differ in terms of engine-related procurement activity and procurement-related lifecycle cost exposure?
- What do the spare-parts records indicate about spare-parts support, stock structure, and maintenance support burden in the two cases?
- How can defect records be used as supporting evidence where available, and what limitations arise when comparable defect data is not available for both vessels?
- What do the available BASSnet records indicate about operational availability, and what limitations arise from the available downtime data?
- How can the results support maintenance strategy decisions for Wilhelmsen Ship Management?

These sub-questions divide the research problem into maintenance activity, procurement-related cost exposure, spare-parts support, defect information, operational availability indicators, data limitations, and practical decision support. Together, they ensure that the comparison does not focus only on contract type, but also considers the operational records, procurement evidence, support structure, and data quality behind each maintenance arrangement.

1.3 Delimitations

This thesis is limited to a comparison of two anonymized vessel cases within the Wilhelmsen Ship Management portfolio. The Passenger Vessel represents the LTSA case, while the Heavy Lift Vessel represents the non-LTSA case. The findings are therefore case-specific and should not be interpreted as universal conclusions for all vessels, engine types, or maintenance contracts.

The two vessels are not identical. They differ in vessel type, operational profile, engine configuration, agreement structure, and data availability. This means that the analysis cannot isolate the effect of LTSA versus non-LTSA alone. Instead, the thesis compares how the two selected maintenance arrangements appear through the available BASSnet records and discusses the limitations that follow from the case differences.

The study focuses on engine-related maintenance and procurement records from BASSnet for the period 2024–2025. The analysis includes job-history records, engine-related purchase-order records, spare-parts records, and defect records where available. Broader vessel-level financial values and Tableau dashboard data are excluded from the final analysis because the available financial values are not sufficiently comparable between the two vessel cases and include broader vessel-level cost structures.

The thesis does not attempt to calculate complete vessel-level cost. Instead, it uses procurement-related cost exposure as an indicator based on available engine-related purchase-order records. This means that the cost analysis reflects visible procurement exposure in the BASSnet data, not the full financial cost of the engines or vessels.

The thesis also does not attempt to create a full engineering simulation of engine degradation or a complete technical availability model. The available BASSnet data provides useful operational evidence, but it does not capture every technical event, every operational consequence, or every downtime event connected to engine maintenance. The analysis is therefore designed as a structured case-based comparison of available operational and procurement records, not as a complete financial or engineering evaluation of the vessels.

Other factors, such as detailed legal contract design, supplier relationship management, crew competence, and organizational capability, are only discussed when they are directly relevant to the interpretation of the results.

1.4 Assumptions

This thesis assumes that the available engine-related BASSnet data from 2024–2025 provides a sufficient basis for comparing the two selected maintenance arrangements at a case-study level. The data is used to compare recorded maintenance activity, procurement activity, spare-parts support, defect information where available, operational availability indicators, and financial risk exposure. However, the data is not assumed to be complete or perfectly symmetrical between the two vessels.

The thesis also assumes that LTSA-based and non-LTSA maintenance arrangements can be compared using the same broad analytical dimensions stated in the problem statement: maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, and financial risk exposure. Since the final analysis is based on BASSnet data, these dimensions are interpreted through operational and procurement-related indicators rather than complete vessel-level financial totals.

A further assumption is that BASSnet records reflect how maintenance and procurement activities were registered in the available system data. Differences in record counts may therefore reflect real operational differences, but they may also reflect differences in registration practice, vessel operation, maintenance structure, agreement structure, or data availability. For this reason, job-history records are not interpreted directly as failure counts, purchase-order rows are not interpreted directly as total cost, and spare-parts records are not interpreted as direct consumption unless this is supported by the available data.

Finally, the thesis assumes that the available records can support a structured maintenance strategy comparison, but not a definitive causal conclusion about whether LTSA or non-LTSA is universally better. The results are therefore interpreted as decision-support evidence for the selected cases, not as a general ranking of all LTSA and non-LTSA maintenance arrangements.

2.0 Literature

This chapter reviews literature relevant to the comparison of LTSA-based and non-LTSA maintenance arrangements for MAN marine engines in an asset-intensive maritime context. The purpose of the chapter is not only to present previous studies, but to show how different research areas help explain the decision problem studied in this thesis. The comparison between LTSA and non-LTSA maintenance is not simply a comparison between two contract types. It is a comparison between two ways of organizing maintenance responsibility, technical support, procurement exposure, spare-parts availability, operational risk, and financial uncertainty.

The literature can therefore be grouped around five connected perspectives. First, service-contract literature explains how full-service and long-term maintenance agreements shift uncertainty and financial exposure between customer and service provider. Second, maritime maintenance literature shows why reliability and availability are critical in vessel operations, especially when engine systems are central to operational continuity. Third, spare-parts logistics literature explains why maintenance performance depends not only on technical planning, but also on whether the right parts can be obtained, delivered, repaired, or replaced at the right time. Fourth, engineering asset-management literature places maintenance decisions within a wider balance between cost, risk, performance, and organizational capability. Finally, uncertainty and quantitative maintenance literature supports the need to consider variation and risk exposure rather than relying only on fixed averages or simple record counts.

Together, these perspectives provide the academic foundation for the thesis. They support the decision to compare the Passenger Vessel and the Heavy Lift Vessel through recorded maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, financial risk exposure, and data quality. At the same time, the literature also shows why the results must be interpreted carefully. The available BASSnet data provides valuable operational evidence, but it does not provide a complete financial, contractual, or technical picture of every consequence connected to LTSA and non-LTSA maintenance.

2.1 Maintenance service contracts and Long-Term Service Agreement

Service-contract literature is central to this thesis because it explains why maintenance arrangements are also risk-allocation mechanisms. Deprez et al. (2021) describe full-service maintenance contracts as agreements where maintenance-related costs are covered over a defined period in exchange for a fixed service fee. This changes the structure of the maintenance decision. Instead of the equipment user carrying uncertain repair and service costs directly, part of this uncertainty is transferred to the service provider. Deprez et al. (2023) develop this further by comparing full-service maintenance contracts with insurance logic, where expected cost depends on both the frequency of maintenance events and the severity of the cost attached to those events.

This frequency-severity perspective is useful for understanding LTSA-based maintenance. An LTSA is not identical to the full-service contracts studied by Deprez et al., but the underlying logic is similar: maintenance uncertainty becomes organized through an agreement that defines responsibility, support, and parts of the financial exposure. In a non-LTSA arrangement, the operator or technical manager may face maintenance events more directly through purchase orders, supplier coordination, spare-parts planning, and refurbishment decisions. In an LTSA-based arrangement, some of these uncertainties may be absorbed, structured, or made more predictable through the agreement.

However, the contract literature also highlights a limitation for this thesis. Studies such as Deprez et al. often rely on detailed contract portfolios, clear maintenance-cost data, and models that estimate expected cost by using historical maintenance interventions and risk factors. The available BASSnet data in this thesis is narrower. It includes job-history records, purchase-order records, spare-parts records, and defect records where available, but it does not provide a complete vessel-level cost picture or a full contractual cash-flow model. For this reason, the service-contract literature is not used as a direct pricing model. Instead, it is used to interpret how LTSA and non-LTSA arrangements may distribute uncertainty differently.

Performance-based contracting adds another relevant perspective. Zhu et al. (2023) show that contract structure can influence maintenance behaviour because payment, responsibility, and performance outcomes are connected. This matters because maintenance contracts do not only allocate cost; they can also shape incentives. If a service provider carries more responsibility for outcomes, it may have stronger incentives to prevent failures, optimize maintenance timing, or improve support processes. If the operator carries more direct responsibility, the organization may have more flexibility and control, but also more exposure to coordination and procurement risk.

At the same time, performance-based contracting should not be treated as the same as the LTSA arrangement studied here. The value of this literature is mainly conceptual. It supports the idea that maintenance arrangements should be analyzed as organizational and financial structures, not only as purchasing categories. This is important for the thesis because the aim is not to prove that LTSA or non-LTSA is universally better, but to examine how the two arrangements appear through available operational and procurement-related evidence.

2.2 Maritime maintenance, reliability, and availability

Maritime maintenance literature shifts the focus from contract structure to operational consequence. In shipping, maintenance decisions are not only about reducing repair cost. They are also about protecting vessel availability, safety, schedule reliability, and long-term asset performance. Kalafatelis et al. (2025) describe how the maritime industry is moving toward more data-driven maintenance approaches because traditional reactive and preventive approaches can be limited when systems are complex and failures are costly. Daya and Lazakis (2024) similarly emphasize that marine machinery maintenance should consider component criticality, failure-prone systems, maintenance delays, and the effect of technical decisions on availability.

This perspective is important because the engines studied in this thesis are not ordinary cost objects. They are critical marine assets. A maintenance event may appear in BASSnet as a job-history record or a purchase-order line, but its operational meaning can be wider than the record itself. A delayed repair, unavailable spare part, or repeated defect can affect operational stability even when the direct cost is not fully visible in the dataset. This is why the thesis includes operational availability indicators and defect information where available, rather than treating procurement exposure as the only relevant outcome.

The maritime maintenance literature also shows a methodological challenge. Reliability and availability studies often require detailed technical data, such as failure events, downtime duration, component condition, running hours, repair time, and system configuration. The final BASSnet dataset does not provide this level of completeness for both vessels. The Heavy Lift Vessel includes defect records, while comparable defect records are not available for the Passenger Vessel. Downtime fields are also limited and cannot be treated as a complete measure of vessel availability.

This means that the thesis cannot claim to calculate full technical availability. Instead, it uses available records as indicators of maintenance activity and possible interruption exposure. This is a more cautious interpretation, but also a more realistic one. It recognizes that operational systems such as BASSnet can provide useful evidence without being complete reliability databases. The literature therefore supports the inclusion of availability as a relevant dimension, while also confirming the need to avoid overinterpreting missing or incomplete downtime data.

2.3 Spare-parts logistics and support structure

Spare-parts logistics connects the technical and financial sides of the thesis. A maintenance strategy can be well planned in theory, but if the required parts are unavailable, delayed, incorrectly stocked, or difficult to refurbish, the maintenance system remains exposed. Mouschoutzi and Ponis (2022) show that maritime spare-parts logistics is especially complex because vessels are moving assets, supply and demand points are geographically dispersed, delivery windows are narrow, and many actors may be involved in procurement, transport, repair, and reverse logistics. This makes spare-parts support a central part of maritime maintenance performance rather than a secondary administrative activity.

This is directly relevant to the LTSA versus non-LTSA comparison. In a non-LTSA arrangement, the operator or technical manager may carry more direct responsibility for identifying demand, sourcing parts, managing suppliers, coordinating refurbishment, and handling lead-time uncertainty. This may provide flexibility and internal control, but it also creates exposure to procurement delays and high-value spare-parts decisions. In an LTSA-based arrangement, selected support responsibilities may be organized through the agreement, potentially reducing some coordination burden but also making the operator dependent on the terms, coverage, and responsiveness of the service structure.

Wang et al. (2024) add a useful operational perspective by linking spare-parts strategy to support probability, fill rate, and equipment availability. Their work supports the idea that spare-parts decisions affect both cost and readiness. For marine engines, this is especially important because components such as turbochargers, cylinder heads, valves, and other high-value repairable items can create both financial exposure and availability risk. A spare part is therefore not only an inventory item; it is part of the maintenance system's ability to respond.

However, the empirical data in this thesis limits how far this literature can be applied. The available spare-parts records mainly describe stock and support structure. They do not provide complete evidence of actual consumption during the study period. Therefore, spare-parts records are used as indicators of support burden, stock exposure, and maintenance preparedness, not as direct proof that listed parts were consumed in 2024–2025. This distinction is important because treating stock records as consumption records would overstate what the data can prove. The literature strengthens the argument that spare parts matter, but the data requires a cautious interpretation of what spare-parts records actually represent.

2.4 Engineering asset management and maintenance strategy

Engineering asset-management literature provides a broader frame for the thesis by showing that maintenance decisions should balance cost, risk, performance, and organizational capability over the asset lifecycle. More et al. (2024) argue that maintenance decision-making should not rely only on subjective risk assessments or isolated technical considerations, but should include economic and quantitative factors. This fits the problem studied here because the choice between LTSA and non-LTSA maintenance is not only about who performs the work. It is also about how the organization manages cost exposure, technical reliability, spare-parts support, and uncertainty over time.

Goli et al. (2025) further show that maintenance strategy selection depends on system criticality, data availability, operational conditions, and the organization's ability to use different types of maintenance information. Corrective, planned, proactive, and predictive maintenance strategies require different levels of data, infrastructure, and expertise. This is relevant because LTSA and non-LTSA arrangements may influence not only cost responsibility, but also the practical ability to plan, monitor, and respond to maintenance needs.

This literature helps avoid a too-simple interpretation of the thesis results. A higher number of job-history records does not automatically mean worse technical performance. It may reflect more detailed registration, a stronger planned-maintenance structure, different engine configuration, different operational intensity, or agreement-driven reporting practices. Similarly, lower visible procurement exposure does not automatically mean lower total maintenance cost or lower risk. It may simply mean that some costs are hidden within contract structures, recorded differently, or not visible in the available BASSnet extracts.

The asset-management perspective therefore supports the use of multiple indicators. Maintenance activity, purchase-order exposure, spare-parts records, defect information, and availability indicators all show different parts of the decision problem. None of them alone is sufficient. The strength of the thesis is that it treats LTSA and non-LTSA maintenance as asset-management structures rather than as isolated cost categories.

2.5 Uncertainty and quantitative maintenance analysis

Uncertainty is a recurring theme across maintenance literature. Failures do not occur at fixed intervals, repair costs vary, spare-parts lead times change, and operational consequences are not always visible in maintenance records. Cheikh et al. (2025) argue that maintenance strategies should be evaluated not only by expected cost, but also by variability, robustness, and downtime-related effects. This is important because a maintenance strategy with a lower average cost may still be risky if outcomes are highly variable or if a small number of events can create large financial exposure.

This point is especially relevant for the procurement-related part of the thesis. Purchase-order values can be unevenly distributed. A few high-value engine-related orders may dominate visible procurement exposure, while many smaller orders may have limited financial impact. Therefore, comparing only total purchase-order value or average order value would give an incomplete picture. Financial risk exposure must also consider variation, concentration, and the possibility that procurement outcomes differ across periods.

Monte Carlo simulation is relevant in this context because it allows uncertainty to be represented through repeated scenarios rather than a single fixed estimate. In this thesis, the simulation is not used to predict complete vessel lifecycle cost. Instead, it is used more narrowly to explore how visible procurement-related cost exposure may vary when purchase-order frequency and purchase-order value uncertainty are considered. This fits the limitations of the data while still adding analytical depth to the comparison.

At the same time, uncertainty modelling depends on assumptions. The Passenger Vessel and Heavy Lift Vessel differ in vessel type, operational profile, engine configuration, agreement structure, and data availability. These differences cannot be removed by simulation. The quantitative analysis can support interpretation, but it cannot transform the dataset into a controlled experiment. This means that the simulation results should be read as decision-support evidence, not as proof that one maintenance arrangement is technically or financially superior in all situations.

2.6 Research gap and contribution

The reviewed literature shows that the main themes of this thesis are well established individually, but less often combined in one practical comparison. Service-contract research explains how maintenance agreements can transfer risk, influence incentives, and create more predictable cost structures. Maritime maintenance research shows why reliability and availability are critical when technical systems affect vessel operations. Spare-parts logistics research highlights the importance of support structure, lead-time uncertainty, repairable components, and procurement coordination. Engineering asset-management research frames maintenance as a balance between cost, risk, performance, and organizational capability, while quantitative maintenance research shows why uncertainty and variation should be considered alongside expected cost.

The gap is therefore not that these topics are unexplored, but that they are often treated separately. Existing studies commonly assume access to more complete technical, financial, or contractual data than what is available in many real company settings. In practice, maintenance strategy decisions may need to be made using operational system data that is incomplete, unevenly structured, or not fully comparable between assets. This is the situation addressed in this thesis, where LTSA-based and non-LTSA maintenance are compared using available engine-related BASSnet records rather than a complete financial or technical dataset.

The contribution of the thesis is therefore a structured, case-based decision-support comparison under real data limitations. It does not attempt to prove that LTSA or non-LTSA maintenance is universally better. Instead, it applies concepts from service-contract theory, maritime maintenance, spare-parts logistics, asset management, and uncertainty analysis to two anonymized vessel cases in the Wilhelmsen Ship Management portfolio. By comparing maintenance activity, procurement-related cost exposure, spare-parts support, defect information where available, operational availability indicators, financial risk exposure, and data quality together, the thesis provides a more realistic basis for evaluating maintenance arrangements than contract type alone.

This literature foundation also prepares the later chapters. The case description explains the practical setting, the method chapter defines how the BASSnet records are filtered and interpreted, and the modelling chapter translates the literature concepts into indicators, formulas, Monte Carlo simulation, and sensitivity analysis. In this way, the literature does not stand separately from the empirical work; it provides the reasoning behind the thesis framework and explains why the comparison must include operational evidence, procurement exposure, support structure, uncertainty, and data quality.

3.0 Theory

This chapter presents the theoretical foundation used to translate the literature review into the analytical framework of the thesis. While Chapter 2 discussed previous research, this chapter defines the concepts that are applied later in the case description, method, modelling, analysis, and discussion. The purpose is therefore not to repeat the literature, but to clarify how concepts such as maintenance strategy, reliability, availability, spare-parts support, service contracts, risk allocation, and uncertainty are used in the comparison between LTSA-based and non-LTSA maintenance arrangements.

The thesis compares two anonymized vessel cases in the Wilhelmsen Ship Management portfolio. The Passenger Vessel represents the LTSA-based case, while the Heavy Lift Vessel represents the non-LTSA case. The comparison is based on available engine-related BASSnet records from 2024–2025, including job-history records, purchase-order records, spare-parts records, and defect records where available. Since these records come from an operational maintenance and procurement system, the theory must support careful interpretation of what the data can and cannot show.

The theoretical framework is built around five analytical dimensions. First, maintenance strategy explains how maintenance responsibility and planning are organized. Second, reliability and availability explain why technical performance matters beyond cost alone. Third, spare-parts support explains how stock structure, parts availability, and procurement coordination affect maintenance responsiveness. Fourth, service-contract theory explains how LTSA-based and non-LTSA arrangements can allocate responsibility and financial exposure differently. Finally, uncertainty theory explains why the comparison must consider variation, data quality, and modelling limitations rather than relying only on simple record counts or fixed averages.

The theory is applied as a decision-support framework, not as a basis for claiming that the two vessels are technically identical. The Passenger Vessel has four engines, configured as two 12-cylinder V-engines and two 9-cylinder inline engines, while the Heavy Lift Vessel has four 12-cylinder V-engines. The cases also differ in vessel type, operational profile, agreement structure, and data availability. These differences mean that the thesis cannot isolate the effect of LTSA versus non-LTSA alone. Instead, the theory supports a structured comparison of how the two arrangements appear through the available BASSnet indicators.

3.1 Engineering asset management and maintenance strategy

Engineering asset management provides the broad decision-making frame for the thesis. In an asset-intensive maritime context, maintenance is not only a technical activity performed when equipment requires service. It is part of a wider asset-management decision where cost, risk, performance, support capability, and operational continuity must be balanced over time. More et al. (2024) emphasize that maintenance decision-making should consider economic and performance-related factors rather than relying only on subjective risk assessments or isolated technical judgments.

This perspective is important because the choice between LTSA-based and non-LTSA maintenance is not simply a question of which arrangement has the lowest visible cost. The selected maintenance arrangement affects who carries responsibility, how maintenance is planned, how spare parts are secured, how procurement exposure appears, and how uncertainty is handled. An LTSA-based arrangement may provide more structured external support and clearer allocation of selected responsibilities, while a non-LTSA arrangement may provide more direct control but also greater exposure to coordination, procurement, and support risk.

Maintenance strategy explains how an organization decides when and how maintenance actions should be performed. Goli et al. (2025) distinguish between corrective, planned, proactive, and predictive maintenance strategies. Corrective maintenance reacts after failure, planned maintenance follows predefined intervals, proactive maintenance responds to condition indicators, and predictive maintenance uses data to anticipate future failures. These categories are useful because they show that maintenance strategy depends not only on technical need, but also on data availability, system criticality, organizational capability, and decision-making structure.

In this thesis, maintenance strategy is measured indirectly through available BASSnet records. Job-history records are used as indicators of recorded maintenance activity. They can show the scale and structure of registered maintenance work, but they are not treated as direct failure counts. A high number of job-history records may reflect more maintenance workload, but it may also reflect stronger documentation, more planned maintenance, different engine configuration, or different registration practice.

The basic maintenance activity indicator used later in the modelling chapter can be expressed as:

$$\text{Maintenance activity rate} = \frac{N_{\text{jobs}}}{T}$$

Here, N_{jobs}represents the number of recorded engine-related job-history records, while T represents the study period. This indicator does not show failure probability. It shows how much maintenance activity was recorded relative to the selected period. This distinction is important because the thesis compares maintenance arrangements through observable system records, not through a complete technical failure model.

3.2 Reliability, availability, maintainability, and dependability

Reliability, availability, maintainability, and dependability are central concepts in maintenance theory. Reliability refers to the probability that a system performs its intended function without failure over a given period. Availability concerns whether the system is operational and ready for use when required. Maintainability describes how quickly and effectively the system can be restored after a failure or maintenance event. Dependability is a broader concept that reflects the overall trustworthiness of system performance.

These concepts matter because marine engines are critical technical assets. A maintenance event is not important only because it creates cost. It can also affect vessel readiness, schedule stability, safety, and operational flexibility. Daya and Lazakis (2024) show that marine machinery reliability is influenced by component criticality, failure-prone systems, maintenance delays, spare-parts availability, and operating conditions. This supports the thesis position that maintenance arrangements should be assessed through both procurement exposure and operational indicators.

In a complete reliability study, availability would normally require detailed information about operating time, downtime, repair time, failure events, and restoration time. The available BASSnet data does not provide this level of completeness for both vessels. The Heavy Lift Vessel has defect records, while the Passenger Vessel does not have a comparable defect dataset in the final data foundation. The downtime fields in the job-history records are also limited. For this reason, the thesis uses availability only as a simplified proxy based on available recorded downtime, not as a full technical availability calculation.

The simplified availability proxy used later can be expressed as:

$$A_{\text{proxy}} = \frac{T - D_{\text{recorded}}}{T}$$

Here, T represents the total study-period time, and D_{recorded}represents downtime recorded in the available BASSnet fields. If recorded downtime is zero or missing, the result should not automatically be interpreted as proof of perfect technical availability. It may instead show that the available records do not capture downtime in a way that supports a complete availability calculation. This is why availability is treated as an indicator and data-quality observation rather than as a definitive measure of operational uptime.

3.3 Spare-parts logistics and support structure

Spare-parts support is a key link between maintenance planning and operational performance. Maintenance cannot be completed effectively if the necessary parts are unavailable, delayed, wrongly stocked, or difficult to repair. In maritime operations, this challenge is stronger because vessels are moving assets, suppliers are geographically dispersed, delivery windows are limited, and repairable components may require reverse logistics. Mouschoutzi and Ponis (2022) emphasize that maritime spare-parts logistics is closely connected to both cost control and asset availability.

This is directly connected to the LTSA versus non-LTSA comparison. Under a non-LTSA arrangement, the operator or technical manager may carry more direct responsibility for identifying parts demand, sourcing suppliers, managing lead times, coordinating refurbishment, and handling procurement exposure. Under an LTSA-based arrangement, some support responsibilities may be structured through the agreement with the manufacturer or service provider. The difference between the arrangements is therefore not only about who pays for maintenance. It is also about how the support system around the engine is organized.

Wang et al. (2024) show that spare-parts strategy can influence equipment availability and spare-parts requirements. This is important for high-value engine components, where parts availability can affect both downtime risk and financial exposure.

A spare part should therefore be understood as more than an inventory item. It is part of the maintenance system's ability to respond when technical work is required.

In the thesis, spare-parts records are interpreted as stock and support-structure indicators. They may include information such as stock quantity, average price, material description, supplier, and criticality-related fields where available. However, a spare-parts record does not necessarily mean that the part was consumed during the study period. The records are therefore not used as direct consumption evidence unless supported by other available data.

The basic stock-value logic used later can be expressed as:

$$V_{\text{stock}} = \sum_{j=1}^{m} q_j \times p_j$$

Here, q_j represents the stock quantity of spare part j, $p_j$represents its average price, and m represents the number of spare-parts records included. This calculation supports analysis of spare-parts exposure and support structure, but it should not be interpreted as actual consumption cost. It shows the visible stock-related value in the available records.

3.4 Service contracts and risk allocation

Service contract theory explains how maintenance arrangements can shift responsibility and financial exposure between the customer, operator, technical manager, and service provider. Deprez et al. (2021) describe full-service maintenance contracts as arrangements where the customer pays a fixed fee or premium in exchange for coverage of maintenance-related costs over a defined period. Deprez et al. (2023) further connect maintenance contract risk to frequency and severity logic, where total exposure depends both on how often maintenance events occur and how costly those events become.

This logic is useful for understanding LTSA-based maintenance. An LTSA is not treated here as identical to a full-service contract in the strict pricing sense, but it can be understood as a structure that organizes responsibility, support, and parts of the uncertainty around future maintenance needs. In a non-LTSA arrangement, procurement and repair exposure may appear more directly in purchase orders and internal coordination work. In an LTSA-based arrangement, some exposure may be structured through recurring agreement-related entries, service-provider involvement, or contract-defined support.

The thesis therefore uses purchase-order records as indicators of procurement-related cost exposure. These records do not represent complete vessel-level cost, and they do not capture all financial effects of the maintenance arrangement. However, they show visible engine-related procurement activity in the available BASSnet data. The core procurement exposure measure can be expressed as:

$$E_{\mathrm{PO}} = \sum_{i=1}^{n} C_i$$

Here, E_{PO}represents total visible procurement-related exposure, $C_i$represents the value of purchase-order entry i, and $n$represents the number of included purchase-order entries after filtering and correction. This measure is useful for comparing visible procurement exposure, but it is not a complete measure of total maintenance cost.

Average purchase-order value can be expressed as:

$$\overline{C_{\mathrm{PO}}} = \frac{E_{\mathrm{PO}}}{n}$$

This average provides a simple cost-exposure indicator, but it must be interpreted together with variation and concentration. A case may have a moderate average value but still contain a few very large orders that dominate total exposure.

Performance-based contracting also adds a useful theoretical perspective. Zhu et al. (2023) show that contract form can influence incentives and maintenance decisions because payment, responsibility, and performance outcomes are connected. Although LTSA is not identical to performance-based contracting, the principle is relevant: contract structure can influence behavior. This supports the thesis argument that LTSA and non-LTSA should be compared as maintenance arrangements with different responsibility and risk structures, not only as procurement categories.

3.5 Risk, uncertainty, and Monte Carlo simulation

Maintenance decisions are made under uncertainty. Engine failures do not occur at perfectly regular intervals, repair needs vary, spare-parts lead times can change, and procurement values may be unevenly distributed. A comparison based only on total cost or average value would therefore give an incomplete picture of the decision problem. Cheikh et al. (2025) support the idea that maintenance strategies should be assessed not only by expected cost, but also by variability, robustness, and downtime-related effects.

In this thesis, uncertainty is especially important in the procurement-related analysis. Purchase-order values can be concentrated, meaning that a small number of high-value orders may account for a large share of total exposure. This matters because financial risk is not only about how much is spent on average. It is also about how unpredictable or concentrated the exposure is.

Uncertainty is also connected to data quality. BASSnet is designed for maintenance planning, procurement, and technical follow-up, not primarily for academic analysis. Job-history records, purchase-order rows, spare-parts records, and defect records may therefore reflect both real operational differences and differences in registration practice. This means that the thesis must combine numerical comparison with careful interpretation.

Monte Carlo simulation provides a way to represent uncertainty through repeated scenarios rather than one fixed estimate. In this thesis, the simulation is used to explore procurement-related cost exposure, not to predict complete vessel lifecycle cost. The simulation uses available purchase-order values to examine how exposure may vary when procurement frequency and purchase-order value uncertainty are considered.

The simplified Monte Carlo exposure logic can be expressed as:

$$E_{\mathrm{PO}}^{(s)} = E_{\mathrm{recurring}} + \sum_{i=1}^{N_s} X_i$$

Here, $E_{\mathrm{PO}}^{(s)}$represents simulated procurement exposure in simulation run s, $E_{\mathrm{recurring}}$represents recurring agreement-related exposure where applicable, $N_s$represents the simulated number of ordinary purchase-order events, and $X_i$represents simulated purchase-order values. This formulation reflects the thesis logic that recurring LTSA-related entries should not be treated the same way as ordinary uncertain purchase orders when the records indicate distinct recurring exposure.

The purpose of the simulation is not to create an exact forecast. It is to show how procurement-related exposure may vary under uncertainty. This is useful because two cases may appear similar under average values but differ in risk profile, concentration, or exposure to high-value procurement events.

Sensitivity analysis is used to test how dependent the results are on assumptions. This is important because the two vessel cases differ in vessel type, engine configuration, operational profile, agreement structure, and data availability. The sensitivity logic can be expressed as:

$$\Delta Y = \frac{Y_{\mathrm{changed}} - Y_{\mathrm{base}}}{Y_{\mathrm{base}}}$$

Here, Y_{base}represents the original model result and Y_{changed}represents the result after changing an assumption. This helps assess whether the interpretation remains stable when uncertain assumptions are adjusted. In the thesis, sensitivity analysis strengthens the comparison by showing whether conclusions depend heavily on specific modelling choices.

3.6 Theoretical framework of the thesis

The theoretical framework combines the concepts above into one decision-support structure. The thesis does not evaluate LTSA and non-LTSA maintenance only by asking which arrangement has more records or lower visible cost. Instead, it compares the two cases through connected indicators that reflect different parts of the maintenance decision.

Engineering asset management provides the overall decision logic: maintenance arrangements should be assessed through cost, risk, performance, and organizational capability. Maintenance strategy explains why job-history records can indicate recorded workload, but not direct failure frequency. Reliability and availability explain why operational indicators matter, while also showing why incomplete downtime data must be interpreted cautiously. Spare-parts theory explains why stock structure and parts support affect responsiveness and downtime risk. Service-contract theory explains why LTSA and non-LTSA arrangements may allocate responsibility and financial exposure differently. Uncertainty theory explains why variation, concentration, Monte Carlo simulation, and sensitivity analysis are needed alongside deterministic indicators.

This framework prepares the later chapters. The method chapter explains how the BASSnet records are filtered and interpreted. The modelling chapter translates the theoretical dimensions into measurable indicators and formulas. The analysis and results chapters then compare how the LTSA-based and non-LTSA cases appear in the available data. Because the data is incomplete and the vessels are not identical, the framework supports a cautious case-based comparison rather than a universal conclusion that one maintenance arrangement is always better than the other.

4.0 Case Description

This chapter presents the practical case used in the thesis. The purpose is to explain the company context, the maintenance situation, the selected vessel cases, and why the comparison between LTSA-based and non-LTSA maintenance arrangements is relevant. The case is based on marine engine maintenance within the Wilhelmsen Ship Management portfolio and uses available engine-related BASSnet records as the empirical foundation.

Wilhelmsen Ship Management is relevant to the case because technical ship management involves maintenance planning, repair coordination, spare-parts support, procurement follow-up, and long-term asset-related decisions. These activities are directly connected to the problem statement of the thesis, which compares maintenance arrangements through maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, and financial risk exposure.

The thesis compares two anonymized vessel cases with different maintenance arrangements. The first vessel is referred to as the Passenger Vessel and represents the LTSA-based case. The second vessel is referred to as the Heavy Lift Vessel and represents the non-LTSA case. Under an LTSA-based arrangement, selected maintenance responsibilities, technical support, spare-parts support, and parts of the cost structure are governed through an agreement with the engine manufacturer or service provider. Under a non-LTSA arrangement, the operator or technical manager carries more direct responsibility for maintenance coordination, spare-parts planning, refurbishment decisions, procurement activity, and cost exposure.

The two cases therefore represent different ways of organizing maintenance responsibility, technical support, procurement exposure, and financial risk. They are not treated as identical vessels. The Passenger Vessel and the Heavy Lift Vessel differ in vessel type, operational profile, engine configuration, agreement structure, and data availability. For this reason, the comparison is interpreted as a case-based analysis of available engine-related BASSnet records, not as a direct one-to-one comparison of identical vessels.

4.1 Maintenance context

The case concerns MAN marine engines used in asset-intensive vessel operations. These engines are mission-critical because propulsion and onboard power availability directly affect vessel operations, service continuity, maintenance planning, and operational reliability. If engine-related failures or maintenance delays occur, the consequences can extend beyond the immediate repair activity and create wider operational and financial effects.

The two vessel cases also differ in engine configuration. The Passenger Vessel has four engines, configured as 2×12V engines and 2×9L inline engines, while the Heavy Lift Vessel has 4×12V engines. This difference is important because engine configuration can influence maintenance workload, spare-parts requirements, procurement activity, and comparability between the cases. The thesis therefore does not assume that differences in BASSnet records are caused only by the maintenance arrangement. They may also reflect differences in vessel type, engine configuration, operation, registration practice, or available data.

The company notes indicate that engine operation and maintenance are strongly linked to running hours. Engines are normally managed so that usage is distributed between comparable units in order to keep them in similar condition and make maintenance planning easier. Maintenance activities are planned through the BASSnet planned maintenance system (PMS), with regular checks and planned maintenance intervals. Based on the operational notes, maintenance and overhaul activities can be connected to intervals such as approximately 6,000, 12,000, 18,000, 24,000, 30,000, and 36,000 running hours, where larger overhauls create greater planning and resource requirements.

The case also involves critical spare parts and repairable components. Company notes identify components such as turbochargers, cylinder heads, and valves as important because they are high-value items and may take longer to obtain, repair, or refurbish. This makes spare-parts planning and procurement important for both cost control and operational availability. In a non-LTSA arrangement, this responsibility is handled more directly by the operator or technical manager, while an LTSA-based arrangement may provide more structured manufacturer support for selected maintenance activities and parts.

4.2 Decision situation

The central decision situation is how engine maintenance should be organized and evaluated when comparing an LTSA-based arrangement with a non-LTSA arrangement. An LTSA-based arrangement may provide more predictable support structures, manufacturer involvement, technical follow-up, and clearer allocation of selected responsibilities. However, it may also involve fixed contractual commitments and less flexibility. A non-LTSA arrangement may provide more direct control over purchasing, refurbishment, and maintenance planning, but it can also expose the operator or technical manager to greater uncertainty in spare-parts availability, procurement lead times, repair needs, and downtime consequences.

This decision cannot be evaluated only by comparing short-term cost indicators. A case with lower visible procurement activity may still involve higher uncertainty, while a case with more recorded maintenance activity may reflect better documentation, different registration practices, or a broader planned maintenance structure. The relevant question is therefore how the two arrangements compare when maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, financial risk exposure, and data quality are considered together.

4.3 Data context

The case is supported by engine-related operational and procurement data from BASSnet. BASSnet provides records such as job history, purchase orders, spare-parts information, and defect records where available. These records make it possible to compare the two vessel cases using actual company system data rather than only theoretical assumptions.

At the same time, the case has important data limitations. The available data is not perfectly symmetrical between the vessels. The Heavy Lift Vessel has defect records, while the Passenger Vessel does not have a comparable defect dataset in the available files. The number and type of job-history records, purchase-order rows, and spare-parts records also differ between the cases. These differences may reflect real operational differences, but they may also reflect differences in vessel type, engine configuration, system use, registration practice, agreement structure, or data availability.

For this reason, the analysis must be interpreted as a case-based decision-support study rather than a complete engineering or accounting comparison of all LTSA and non-LTSA situations. Job-history records are used as indicators of recorded maintenance activity, purchase-order records are used as indicators of procurement-related cost exposure, spare-parts records are used to understand support structure, and defect records are used only where available.

4.4 Relevance of the case

This case is relevant because it combines the main elements of the thesis problem: critical marine engines, uncertain maintenance needs, spare-parts dependency, procurement activity, contract structure, operational availability, financial risk exposure, and data quality. It also reflects a real decision problem in asset-intensive maritime operations, where companies must balance predictability, flexibility, control, support structure, and maintenance-related uncertainty.

The case therefore provides the practical foundation for the analysis in the following chapters. The Passenger Vessel and the Heavy Lift Vessel are used to compare how LTSA-based and non-LTSA maintenance arrangements appear under the available BASSnet data and stated assumptions. The aim is not to prove that one arrangement is universally better, but to support maintenance strategy decision-making for Wilhelmsen Ship Management using available operational and procurement-related evidence.

5.0 Method and Data

This chapter describes the methodological approach and data foundation used in the thesis. The study uses a quantitative comparative case-study design to compare two maintenance arrangements for MAN marine engines within the Wilhelmsen Ship Management portfolio. The Passenger Vessel represents the LTSA-based case, while the Heavy Lift Vessel represents the non-LTSA case. The comparison is based on available engine-related BASSnet records from the period 2024–2025.

The purpose of the method is to structure available company data into a form that can support comparison of maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, and financial risk exposure. The data is not used to create a complete engineering simulation of the engine systems or a complete accounting model of vessel costs. Instead, it is used to build a transparent case-based comparison based on the available BASSnet records.

The final data foundation consists of CSV files exported from BASSnet. These files include job-history records, engine-related purchase-order records, spare-parts records, and defect records where available. Since the data comes from operational company systems, it must be interpreted carefully. The records are useful for comparing maintenance activity, procurement activity, spare-parts support, and availability-related indicators, but they do not provide a complete picture of all technical events, all operational consequences, or all financial effects connected to engine maintenance.

5.1 Method

The thesis applies a quantitative comparative case-study method. A case-study approach is suitable because the research problem is linked to a real maintenance strategy decision context involving specific vessels, company systems, engine configurations, and maintenance arrangements. The comparison is not intended to represent all vessels or all LTSA-based and non-LTSA arrangements. Instead, it examines how the two selected cases differ when analyzed through available engine-related BASSnet data.

The study is mainly descriptive and comparative. It is descriptive because it organizes and explains recorded maintenance and procurement data from BASSnet. It is comparative because the same main categories of information are structured for both vessel cases where possible. These categories include job history, purchase orders, spare parts, and defect records where available.

The empirical data was not collected through surveys or formal interviews. Instead, the thesis uses system-based records from BASSnet. BASSnet provides operational and procurement-related information such as job-history records, purchase orders, spare-parts records, and defect records where available. Practical company notes are used as contextual support, especially for understanding the maintenance context, engine configuration, and the role of running-hour-based maintenance planning.

The method followed a structured process. First, the available CSV files were separated by vessel case and data type. Second, the data was restricted to the 2024–2025 study period. Third, the records were filtered to focus on engine-related maintenance and procurement activity. Fourth, the data was organized into comparable categories such as job history, purchase orders, spare parts, and defects. Finally, the records were interpreted using defined rules so that different record types were not treated as if they represented the same type of evidence.

Because the data comes from operational company systems, it must be interpreted with caution. BASSnet is designed for maintenance planning, procurement, and technical follow-up, not primarily for academic research. Some fields may be incomplete, repeated, inconsistently registered, or not directly comparable between vessels. These limitations are handled by documenting the data interpretation rules and by treating the results as a case-based comparison rather than as a complete technical or financial evaluation.

5.2 Data

The thesis uses engine-related BASSnet CSV files prepared for the 2024–2025 period. The files contain maintenance and procurement-related records for the Passenger Vessel and the Heavy Lift Vessel. The main datasets are summarized in Table 5.1.

Table 5.1: Main BASSnet CSV data sources used in the thesis

Data source	Vessel case	Strategy	Main content	Period	Records	Use in thesis
Job-history CSV	Passenger Vessel	LTSA	Engine-related job-history records	2024	1,454	Maintenance activity
Job-history CSV	Passenger Vessel	LTSA	Engine-related job-history records	2025	1,237	Maintenance activity
Purchase-order CSV	Passenger Vessel	LTSA	Engine-related purchase-order records	2024-2025	134 rows	Procurement activity and cost exposure indicators
Spare-parts CSV	Passenger Vessel	LTSA	Spare-parts master and stock-related records	2024-2025	104	Spare-parts support structure
Defect CSV	Heavy Lift Vessel	Non-LTSA	Defect records	2024-2025	99	Supporting defect evidence
Job-history CSV	Heavy Lift Vessel	Non-LTSA	Engine-related job-history records	2024-2025	1,062	Maintenance activity
Purchase-order CSV	Heavy Lift Vessel	Non-LTSA	Engine-related purchase-order records	2024-2025	83 rows	Procurement activity and cost exposure indicators
Spare-parts CSV	Heavy Lift Vessel	Non-LTSA	Spare-parts master and stock-related records	2024-2025	121	Spare-parts support structure

Source: Prepared by Evan Elias Håpnes based on BASSnet CSV exports from Wilhelmsen Ship Management.

The Passenger Vessel has a total of 2,691 job-history records across the 2024–2025 period. This consists of 1,454 records for 2024 and 1,237 records for 2025. The Passenger Vessel also has 134 purchase-order rows and 104 spare-parts records. The Heavy Lift Vessel has 1,062 job-history records for the 2024–2025 period. It also has 83 engine-related purchase-order rows, 121 spare-parts records, and 99 defect records.

The difference in defect data is important. The Heavy Lift Vessel has defect records, while the Passenger Vessel does not have a comparable defect dataset in the available CSV files. This does not mean that the Passenger Vessel had no defects during the period. It only means that comparable defect data was not available in the final data foundation. For this reason, defect-based conclusions must be treated carefully and cannot be used as a direct defect-frequency comparison between the two cases.

5.3 Data filtering and preparation

The BASSnet data was filtered to focus on engine-related records. For the job-history data, the filtering was based on vessel-specific engine-system categories available in the BASSnet export. The Passenger Vessel and the Heavy Lift Vessel use different internal category labels for their engine-related systems. For confidentiality reasons, these internal category labels are not reproduced in the thesis. The purpose of the filter was to retain records related to the relevant main engine systems while excluding unrelated vessel systems.

A broader machinery-code filter was also considered during the data preparation process. However, the final filtering approach was selected based on which method gave the most relevant engine-related records without including unrelated technical systems. This means that the job-history data was not filtered only by a general vessel-wide machinery category, but by the engine-relevant categories that best matched the maintenance scope of the thesis.

For procurement data, the filtering was based on engine-related purchase-order content, supplier information, account classification, and descriptions where available. The aim was to retain purchase-order records connected to engine maintenance, engine-related parts, service support, LTSA-related activity, and relevant procurement exposure. Internal account codes and category labels are not reproduced in the main text because the thesis uses anonymized vessel cases. Instead, the method describes the filtering logic and interprets the results at an aggregated case level.

The purchase-order data required additional preparation because purchase-order rows are not always equivalent to unique purchase orders. Some rows may represent line items, updates, versions, or repeated registrations. For ordinary purchase-order version updates, the latest non-cancelled version is used to reduce double-counting. However, recurring LTSA-related monthly or running-hour-based rows are treated separately when the descriptions and registration pattern indicate that they represent distinct recurring cost exposure entries rather than duplicate versions of the same order. This distinction is important because treating all recurring LTSA rows as duplicate purchase-order versions would understate the visible LTSA-related procurement exposure.

Cancelled purchase orders are excluded from cost-exposure calculations where the purpose is to estimate active procurement exposure. Monetary fields are cleaned and interpreted consistently, including handling values with thousands separators and decimal notation. All purchase-order values used in the analysis are treated as indicators of visible procurement-related cost exposure in BASSnet, not as a complete financial statement of vessel costs.

Spare-parts data was prepared by organizing the available spare-parts records by vessel case and relevant stock-related fields. These records include information such as material descriptions, stock quantities, average prices, suppliers, and other support-structure information where available. Spare-parts records are not treated as direct consumption records unless another record supports that interpretation.

Defect data was used only where available. The Heavy Lift Vessel has a defect dataset for the selected period, while the Passenger Vessel does not have a comparable defect file in the available data foundation. The defect data is therefore used as supporting evidence for the Heavy Lift Vessel case, not as a direct defect-frequency comparison between the two vessel cases.

5.4 Data interpretation rules

The four main data categories are interpreted differently because they represent different types of operational evidence.

Job-history records are used as indicators of recorded maintenance activity. A job-history record may represent planned maintenance, inspection work, corrective work, administrative follow-up, or other registered maintenance activity. Therefore, job-history records are not interpreted directly as failure counts. A higher number of job-history records may reflect more maintenance activity, but it may also reflect differences in vessel type, engine configuration, planned maintenance structure, registration practice, agreement structure, or system use.

Purchase-order records are used as indicators of procurement activity and procurement-related cost exposure. A purchase-order row is not always the same as a unique purchase order because the same purchase order may appear in several rows due to versions, line items, updates, or registrations. The analysis therefore separates ordinary purchase-order version handling from recurring LTSA-related cost exposure. For ordinary repeated versions, the latest non-cancelled version is used. For recurring LTSA-related monthly or running-hour-based entries, the rows are treated as separate exposure entries when the descriptions indicate distinct recurring costs. These records are used to understand procurement activity, supplier interaction, agreement-related exposure, and maintenance-related purchasing, but they are not treated as a complete financial cost statement.

Spare-parts records are treated as master, stock, and support-structure data. They describe information such as material names, stock levels, average prices, suppliers, criticality, and lead-time-related information where available. These records are useful for understanding spare-parts support burden and inventory structure. However, they should not be interpreted as direct evidence that the listed parts were consumed during the 2024–2025 period unless this is supported by another record.

Defect records are used only where available. The Heavy Lift Vessel has 99 defect records in the available data, while the Passenger Vessel does not have a comparable defect file. The defect data can therefore support the interpretation of the Heavy Lift Vessel case, but it cannot be used to compare defect frequency directly between the two vessels.

5.5 Data preparation and limitations

The data preparation process focused on making the available BASSnet CSV records usable for comparison. The datasets were organized by vessel case, strategy, year, and data type. The 2024–2025 period was used as the common time frame for both cases. Job-history records were used to describe maintenance activity, purchase-order records were used to describe procurement activity and procurement-related cost exposure, spare-parts records were used to describe support structure, and defect records were used where available.

Several limitations affect the analysis. The most important limitation is that the two vessel cases do not have identical data availability. The Heavy Lift Vessel has defect data, while the Passenger Vessel does not have a comparable defect dataset in the final CSV files. This limits the precision of direct defect-based comparison.

Another limitation is that the two vessels are not identical. The Passenger Vessel and Heavy Lift Vessel differ in vessel type, operational profile, engine configuration, agreement structure, and maintenance context. Differences in record counts may therefore reflect real operational differences, but they may also reflect differences in how work is planned, registered, and followed up in BASSnet.

The purchase-order data must also be interpreted carefully. Purchase-order rows may represent versions, line items, recurring monthly agreement entries, or ordinary procurement activity. The values in the purchase-order files can support analysis of procurement-related cost exposure, but they should not be treated as a complete accounting measure of total vessel cost. The method therefore distinguishes between ordinary purchase-order version updates and recurring LTSA-related cost exposure entries. This reduces the risk of both double-counting ordinary purchase-order updates and undercounting recurring LTSA-related costs.

Similarly, spare-parts records provide useful information about support structure and stock characteristics, but they are not direct consumption records. A part listed in the spare-parts file may represent stock structure or master-data registration rather than actual use during the selected period.

The available data also does not provide a complete engineering measure of vessel availability. Job-history and defect records can indicate maintenance activity and possible interruption exposure, but they do not capture every operational consequence or every downtime event. Any availability-related conclusion in the thesis should therefore be interpreted as an operational indicator based on available BASSnet data, not as a complete technical availability calculation.

These limitations do not make the data unusable, but they define how the results should be interpreted. The thesis therefore presents the findings as a structured case-based comparison of the Passenger Vessel and the Heavy Lift Vessel, not as a universal conclusion about all LTSA-based and non-LTSA maintenance arrangements.

5.6 Use of AI tools

Artificial intelligence tools, including ChatGPT, were used as support during the preparation of this thesis. The AI tools were used for programming assistance, debugging, structuring Python code, explaining and formatting mathematical formulas, supporting calculation logic, supporting Monte Carlo and sensitivity analysis, assisting with digital data analysis, and improving clarity in selected parts of the text.

This applies especially to the mathematical and modelling-related parts of the thesis, including the modelling framework, procurement-exposure calculations, Monte Carlo simulation, sensitivity analysis, and the Python code appendix. The AI tools were not used as independent academic sources. All formulas, code, calculations, model assumptions, results, interpretations, and conclusions were reviewed, tested, edited, and approved by Evan Elias Håpnes. The final thesis is the author's own work, and responsibility for the submitted content remains with the author.

6.0 Modelling

This chapter presents the modelling framework used to compare the LTSA-based and non-LTSA maintenance arrangements. The model is designed as a simplified quantitative decision-support model, not as a complete engineering simulation of the engine systems and not as a full accounting model of vessel cost. Its purpose is to connect the available engine-related BASSnet CSV data to the thesis problem statement by comparing the two cases in terms of maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, and financial risk exposure.

The Passenger Vessel represents the LTSA-based case, while the Heavy Lift Vessel represents the non-LTSA case. The model uses the 2024–2025 study period as the analysis horizon. This period consists of 731 calendar days, equal to 17,544 hours if continuous operation is assumed. The results should therefore be interpreted as a case-based comparison of two selected vessels, not as a universal conclusion about all LTSA-based and non-LTSA maintenance arrangements.

The modelling formulas, calculation logic, Monte Carlo simulation, sensitivity analysis, and related Python code were developed with AI assistance for explanation, formatting, programming support, and verification of calculation structure. The final formulas, assumptions, calculations, code outputs, interpretations, and conclusions were reviewed, tested, edited, and approved by Evan Elias Håpnes.

The model is based on four main categories of BASSnet data: job-history records, purchase-order records, spare-parts records, and defect records where available. These data categories are not treated as identical types of evidence. Job-history records are used to indicate recorded maintenance activity, purchase-order records are used to estimate procurement-related cost exposure, spare-parts records are used to describe stock and support structure, and defect records are used only where available.

All monetary values in the model are presented in USD, based on the currency used in the final BASSnet CSV files. Non-monetary values, such as job-history records, purchase-order counts, spare-parts counts, stock quantities, and downtime hours, are presented as counts or hours.

The modelling follows a structured process. First, the BASSnet CSV files are prepared and filtered. Second, deterministic indicators are calculated from the observed records. Third, purchase-order exposure is corrected by separating ordinary purchase-order version updates from recurring LTSA-related entries. Fourth, Monte Carlo simulation is used to represent uncertainty in procurement-related cost exposure. Fifth, sensitivity analysis is used to test how robust the comparison is when key assumptions are changed.

6.1 Model objective and scope

The objective of the model is to evaluate how the LTSA-based and non-LTSA cases differ across the main dimensions in the problem statement: maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, and financial risk exposure. Since the final dataset consists of BASSnet operational and procurement records, these dimensions are interpreted through indicators rather than complete vessel-level financial totals.

Maintenance activity is represented through job-history records. Procurement-related cost exposure is represented through engine-related purchase-order records. Spare-parts support is represented through spare-parts records, stock quantities, and stock-value indicators. Operational availability is represented through available downtime fields in the job-history records. Financial risk exposure is represented through variation, concentration, and uncertainty in purchase-order values, supported by Monte Carlo simulation and sensitivity analysis.

The model does not include total vessel profitability, total vessel operating cost, or all technical consequences of engine maintenance. It also does not assume that the two vessels are identical. The Passenger Vessel and Heavy Lift Vessel differ in vessel type, operational profile, engine configuration, agreement structure, and data availability. The purpose is therefore not to prove that one arrangement is always better, but to compare how the two selected cases appear under the available BASSnet data and modelling assumptions.

6.2 Model components and assumptions

The model consists of parameters, evaluated alternatives, output measures, and assumptions. The evaluated alternatives are the LTSA-based case represented by the Passenger Vessel and the non-LTSA case represented by the Heavy Lift Vessel.

Component	**Description**
Evaluated alternatives	LTSA-based case represented by the Passenger Vessel, and non-LTSA case represented by the Heavy Lift Vessel
Parameters	BASSnet record counts, purchase-order values, spare-parts stock values, downtime hours, study period, and simulation settings
Output measures	Maintenance activity indicators, procurement-related cost exposure, spare-parts exposure, availability proxy, and financial risk measures
Assumptions	Ordinary purchase-order version updates are handled using the latest non-cancelled version, recurring LTSA-related entries are treated as distinct exposure entries, cancelled orders are excluded from cost-exposure calculations, spare-parts records are not treated as consumption, and downtime fields are interpreted as incomplete availability evidence

The main model input values from the final BASSnet CSV files are summarized in Table 6.1.

Table 6.1: Main model input values from BASSnet CSV files

Input Category	Passenger Vessel, LTSA	Heavy Lift Vessel, non-LTSA
Job-history records	2,691	1,062
Corrective job records, based on Job Class	59	1
Overdue job records	658	105
Recorded downtime hours	0	0
Purchase-order rows	134	83
Ordinary latest-version, non-cancelled purchase orders	50	69
Recurring agreement-related entries retained separately	66	0
LTSA running-hour/monthly exposure, USD	1,359,948.51	Not applicable

Long-lead/Buyersclub recurring exposure, USD	58,332.24	Not applicable
Ordinary purchase-order exposure, USD	755,946.97	454,868.63
Corrected purchase-order exposure indicator, USD	2,174,227.72	454,868.63
Mean corrected purchase-order event value, USD	18,743.34	6,592.30
Standard deviation of corrected purchase-order event value, USD	22,204.15	9,045.92
Coefficient of variation	1.18	1.37
Top-five purchase-order concentration	20.74%	35.91%
Spare-parts records	104	121
Spare-parts records with stock above zero	16	112
Total stock quantity	30	777
Spare-parts stock-value indicator, USD	7,804.87	244,603.78
Defect records	Not available	99

The purchase-order value indicator is corrected compared with a pure latest-version approach. For ordinary purchase orders, the latest non-cancelled version is used to reduce double-counting. However, recurring LTSA-related monthly, running-hour, and long-lead agreement entries are retained as separate exposure entries when the descriptions and registration pattern indicate that they represent distinct recurring costs rather than duplicate versions of the same order. This distinction is necessary because treating all recurring LTSA rows as duplicate purchase-order versions would understate the visible LTSA-related procurement exposure.

6.3 Maintenance activity and availability indicators

Maintenance activity is measured using job-history records. The total number of job-history records for case s is calculated as:

$$J_s = \sum_{y \in \{2024,2025\}} J_{s,y}$$

For the Passenger Vessel, the job-history records are split across two CSV files:

$$J_{Passenger} = 1454 + 1237 = 2691$$

For the Heavy Lift Vessel, the job-history file contains:

$$J_{HeavyLift} = 1062$$

The corrective maintenance share is calculated using records where the Job Class is registered as corrective maintenance:

$$CR_s = \frac{J_s^{corr}}{J_s}$$

Using the BASSnet CSV data:

$$CR_{Passenger} = \frac{59}{2691} = 2.19\%$$

$$CR_{HeavyLift} = \frac{1}{1062} = 0.09\%$$

The overdue job share is calculated as:

$$OR_s = \frac{J_s^{overdue}}{J_s}$$

Using the BASSnet CSV data:

$$OR_{Passenger} = \frac{658}{2691} = 24.45\%$$

$$OR_{HeavyLift} = \frac{105}{1062} = 9.89\%$$

These indicators should not be interpreted as direct failure rates. A job-history record may describe planned maintenance, inspection work, corrective work, cleaning, service, replacement, or administrative follow-up. A higher number of job-history records may reflect more recorded maintenance activity, but it may also reflect differences in vessel type, engine configuration, planned maintenance structure, registration practice, agreement structure, or system use.

Operational availability is included because the problem statement concerns operational availability indicators as well as procurement-related cost exposure and financial risk exposure. However, the available BASSnet data does not provide a complete technical availability record. The model therefore uses a simplified availability proxy based on the recorded downtime field in the job-history records.

At a general level, availability can be expressed as:

$$A = \frac{Uptime}{Uptime + Downtime}$$

For this thesis, the total operating horizon is represented as:

$$T = 731 \times 24 = 17544 \text{ hours}$$

The availability proxy for case s is calculated as:

$$A_s = \frac{T - D_s}{T}$$

In the final BASSnet job-history CSV files, both vessel cases have zero positive recorded downtime hours in the downtime field:
Therefore, the availability proxy becomes:

$$A_{Passenger} = \frac{17544 - 0}{17544} = 100\%$$

$$A_{HeavyLift} = \frac{17544 - 0}{17544} = 100\%$$

This result should not be interpreted as proof that the vessels had perfect operational availability. It only means that the available BASSnet downtime field does not contain positive downtime values for the selected engine-related job-history records. The availability proxy is therefore mainly a data-quality finding and should be discussed as a limitation.

6.4 Procurement and spare-parts cost exposure model

Purchase-order records are used as the main indicator of procurement-related cost exposure in the model. Since the final dataset is based on BASSnet operational and procurement data, the model does not calculate complete vessel-level cost. Instead, it estimates the procurement-related cost exposure visible in the engine-related purchase-order records. All purchase-order values are presented in USD.

The purchase-order exposure indicator for case s is calculated as:

$$V_s^{PO} = R_s + \sum_{p=1}^{U_s^O} V_{p,s}^{latest}$$

Where R_s is the value of recurring agreement-related entries retained as distinct exposure entries, and $V_{p,s}^{latest}$ is the USD value of the latest non-cancelled version of ordinary purchase order p.

For the Passenger Vessel:

$$R_{Passenger} = 1{,}359{,}948.51 + 58{,}332.24 = 1{,}418{,}280.75$$

$$V_{Passenger}^{ordinary} = 755{,}946.97$$

$$V_{Passenger}^{PO} = 1{,}418{,}280.75 + 755{,}946.97 = 2{,}174{,}227.72$$

For the Heavy Lift Vessel, there are no comparable recurring LTSA-related entries in the available purchase-order data:

$$R_{HeavyLift} = 0$$

$$V_{HeavyLift}^{PO} = 454{,}868.63$$

The procurement exposure ratio is calculated as:

$$PER = \frac{V_{Passenger}^{PO}}{V_{HeavyLift}^{PO}}$$

Using the observed values:

$$PER = \frac{2{,}174{,}227.72}{454{,}868.63} = 4.78$$

This means that the Passenger Vessel has approximately 4.78 times higher visible procurement-related cost exposure than the Heavy Lift Vessel in the corrected BASSnet purchase-order indicator. This should not be interpreted as a complete vessel-cost ratio. It only reflects the USD-denominated procurement-related cost exposure visible in the available engine-related BASSnet records.

Spare-parts records are used to describe support structure, stock position, and potential spare-parts exposure. The model does not treat spare-parts records as direct consumption history. Instead, they are interpreted as inventory and support-structure evidence.

The spare-parts stock-value indicator is calculated as:

$$SV_s = \sum_{i=1}^{SP_s} Stock_{i,s} \times AveragePrice_{i,s}$$

Using the BASSnet CSV data:

$$SV_{Passenger} = USD\ 7{,}804.87$$

$$SV_{HeavyLift} = USD\ 244{,}603.78$$

The spare-parts stock-value ratio is calculated as:

$$SPR = \frac{SV_{HeavyLift}}{SV_{Passenger}}$$

Using the observed values:

$$SPR = \frac{244{,}603.78}{7{,}804.87} = 31.34$$

This indicates that the Heavy Lift Vessel has a substantially higher visible spare-parts stock-value indicator in the available BASSnet data. However, this result must be interpreted carefully. Spare-parts records reflect stock and master-data structure, not necessarily actual consumption during the period. Differences may also reflect vessel type, inventory policy, stock registration, or agreement structure.

6.5 Monte Carlo simulation and sensitivity analysis

Monte Carlo simulation is used to represent uncertainty in procurement-related cost exposure. The simulation does not estimate complete vessel cost. Instead, it simulates possible variation in engine-related purchase-order exposure based on the observed BASSnet purchase-order values. All simulated cost-exposure outputs are presented in USD.

The simulation separates recurring agreement-related exposure from ordinary purchase-order exposure. This is important because recurring LTSA-related monthly and running-hour entries are not treated as duplicate purchase-order versions. They are also not treated in the same way as random ordinary purchase orders. Instead, they are included as observed recurring exposure in the Passenger Vessel case, while ordinary purchase orders are simulated as uncertain procurement events.

For each vessel case, the simulated number of ordinary purchase-order events is represented as:

$$N_s^O(m) \sim Poisson(\lambda_s^O)$$

Where $N_s^O(m)$is the simulated number of ordinary purchase-order events for case $s$in simulation run m, and $\lambda_s^O$is the observed number of ordinary latest-version, non-cancelled purchase orders.

For the Passenger Vessel:

$$\lambda^{O}_{Passenger} = 50$$

For the Heavy Lift Vessel:

$$\lambda^{O}_{HeavyLift} = 69$$

The simulated procurement-related cost exposure for case s in simulation run m is calculated as:

$$C_s^{PO}(m) = R_s + \sum_{i=1}^{N_s^O(m)} X_{i,s}^O$$

Where R_s is the recurring agreement-related exposure and $X_{i,s}^O$ is a randomly sampled ordinary purchase-order value from the observed latest-version, non-cancelled ordinary purchase-order values for case s.

For the Passenger Vessel:

$$R_{Passenger} = 1{,}418{,}280.75$$

For the Heavy Lift Vessel:

$$R_{HeavyLift} = 0$$

The expected simulated procurement exposure is calculated as:

$$\widehat{\mu_s} = \frac{1}{M}\sum_{m=1}^{M} C_s^{PO}(m)$$

The simulation uses:

$$M = 10000$$

The model also calculates the median, standard deviation, 5th percentile, 95th percentile, and paired difference between the two vessel cases. These outputs are used later in the results chapter to compare average exposure and financial risk.

Financial risk exposure is measured through variation and concentration in purchase-order values. The coefficient of variation is used to compare relative variation between the two vessel cases:

$$CV_s = \frac{\sigma_s}{\mu_s}$$

Using the corrected purchase-order exposure-event values:

$$CV_{Passenger} = \frac{22{,}204.15}{18{,}743.34} = 1.18$$

$$CV_{HeavyLift} = \frac{9{,}045.92}{6{,}592.30} = 1.37$$

Both cases show high variation in purchase-order values. This means that procurement exposure is not evenly distributed across all purchase orders. A limited number of larger purchase orders can have a large influence on total procurement exposure. The Heavy Lift Vessel has the higher coefficient of variation, while the Passenger Vessel has the higher total visible procurement-related cost exposure.

The top-five purchase-order concentration is calculated as:

$$TC_s = \frac{\sum_{p=1}^{5} V_{p,s}^{top}}{V_s^{PO}}$$

Using the corrected BASSnet CSV data, the top-five concentration is approximately 20.74% for the Passenger Vessel and 35.91% for the Heavy Lift Vessel. This supports the interpretation that the Heavy Lift Vessel has more concentrated procurement exposure, while the Passenger Vessel has higher total exposure spread across more recurring agreement-related entries and ordinary purchase-order records.

Sensitivity analysis is used to test how robust the model results are when important assumptions are changed. This is necessary because the available BASSnet data does not provide a complete financial or technical model. The purpose is therefore not to prove one exact total cost, but to examine how sensitive the comparison is to assumptions about purchase-order exposure, spare-parts stock value, and downtime-related consequences.

The first sensitivity test examines the purchase-order break-even factor. This shows how much the Heavy Lift Vessel's purchase-order exposure would need to increase before it equals the Passenger Vessel's corrected purchase-order exposure:

$$BE_{PO} = \frac{V_{Passenger}^{PO}}{V_{HeavyLift}^{PO}}$$

Using the observed USD values:

$$BE_{PO} = \frac{2{,}174{,}227.72}{454{,}868.63} = 4.78$$

This means that the Heavy Lift Vessel's visible purchase-order exposure would need to increase by approximately 378% before it equals the Passenger Vessel's corrected purchase-order exposure.

The second sensitivity test examines spare-parts stock-value exposure. Spare-parts records are not treated as consumption, but they represent potential inventory and support exposure. A spare-parts weighting factor $\theta$is introduced to test how much of the stock-value indicator should be included in the overall exposure:

$$C_s^{total}(m) = C_s^{PO}(m) + \theta SV_s$$

The spare-parts break-even factor is calculated as:

$$\theta^* = \frac{V_{Passenger}^{PO} - V_{HeavyLift}^{PO}}{SV_{HeavyLift} - SV_{Passenger}}$$

Using the observed USD values:

$$\theta^* = \frac{2{,}174{,}227.72 - 454{,}868.63}{244{,}603.78 - 7{,}804.87} = 7.26$$

This means that the Heavy Lift Vessel's higher spare-parts stock-value indicator would need to be weighted very heavily before it overturns the Passenger Vessel's higher corrected purchase-order exposure. Since spare-parts stock is not the same as spare-parts consumption, this result should be interpreted as a robustness test rather than as a direct cost conclusion.

The third sensitivity test concerns downtime. The BASSnet job-history records show zero positive recorded downtime hours for both vessel cases. This does not prove that there was no actual operational downtime. It only means that the available downtime field does not provide sufficient evidence for a complete availability comparison. Downtime cost exposure can be represented as:

$$C_s^{down} = r_d \times D_s$$

Where r_d is the assumed downtime cost per hour in USD and D_s is the assumed downtime hours. The total exposure including downtime can then be expressed as:

$$C_s^{total}(m) = C_s^{PO}(m) + \theta SV_s + C_s^{down}$$

Because the observed downtime field contains no positive downtime values for either vessel case, downtime is treated only as a scenario variable in the sensitivity analysis. This allows the thesis to discuss how downtime would affect the comparison without claiming that the available BASSnet downtime field provides a complete technical availability measure.

6.6 Model assumptions and limitations

The model provides a structured way to compare the two maintenance arrangements, but it is based on several important limitations. The Passenger Vessel and the Heavy Lift Vessel are different vessel types with different operational profiles, engine configurations, maintenance structures, agreement structures, and data availability. The comparison should therefore be understood as a case-based analysis of two selected vessels, not as evidence that one maintenance arrangement is generally superior in all maritime operations.

The model also depends on how maintenance and procurement activities are registered in BASSnet. Job-history records are used as indicators of recorded maintenance activity, but they are not interpreted as direct failure counts. Purchase-order records are used as indicators of procurement-related cost exposure, but they do not represent a complete financial cost calculation. Spare-parts records are interpreted as stock and support-structure data, not as direct consumption history. This distinction is important because the purpose of the model is to compare visible operational and procurement indicators, not to recreate a full accounting or engineering history of the vessels.

The available data is not perfectly symmetrical between the two vessel cases. The Heavy Lift Vessel has defect-level data, while the Passenger Vessel does not have a comparable defect dataset in the final CSV files. As a result, defect frequency cannot be compared directly between the two cases. Similarly, the downtime field in the job-history records shows zero positive recorded downtime hours for both vessels. This means that the availability proxy is useful mainly as a data-quality observation, not as a complete technical availability measure.

The purchase-order model has been corrected to separate ordinary purchase-order version updates from recurring LTSA-related cost exposure entries. For ordinary purchase orders, the latest non-cancelled version is used to reduce double-counting. For recurring LTSA-related monthly, running-hour, and long-lead agreement entries, the rows are retained as distinct exposure entries when the descriptions and registration pattern indicate separate recurring costs. This improves the model because it avoids both double-counting ordinary purchase-order versions and undercounting LTSA-related recurring exposure.

To improve reproducibility, the model follows a consistent data-cleaning and calculation process. Purchase-order rows are classified into ordinary non-recurring purchase orders and recurring agreement-related entries. Cancelled purchase orders are excluded from cost-exposure calculations. Monetary values are cleaned consistently, including values with thousands separators and decimal notation. Spare-parts stock value is calculated by multiplying stock quantity by average price. The Monte Carlo simulation uses 10,000 iterations and should use a fixed random seed in the final Python appendix so that the results can be reproduced.

Overall, the model should be understood as a simplified but decision-relevant framework. It does not capture the full technical complexity of marine engine maintenance, but it provides a transparent way to connect the available 2024–2025 BASSnet data to the thesis question: how LTSA-based and non-LTSA maintenance arrangements differ in terms of maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, financial risk exposure, and data quality.

7.0 Analysis

This chapter explains how the prepared BASSnet data and model outputs are analyzed before the results are presented. The analysis follows the methodological approach in Chapter 5 and the modelling framework in Chapter 6. The purpose is to evaluate the two anonymized vessel cases across the main dimensions of the thesis: maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, financial risk exposure, and data quality.

The Passenger Vessel is analyzed as the LTSA-based case, while the Heavy Lift Vessel is analyzed as the non-LTSA case. The analysis is based only on the final engine-related BASSnet CSV files for the 2024–2025 period. These files include job-history records, purchase-order records, spare-parts records, and defect records where available. The analysis does not compare total vessel budgets or broader vessel-level financial totals because the two vessels have different vessel types, operational profiles, engine configurations, agreement structures, and data availability.

The chapter is divided into several analytical parts. First, the structure and comparability of the data are assessed. Second, maintenance activity is analyzed using job-history indicators. Third, procurement-related cost exposure is analyzed using corrected purchase-order indicators that separate ordinary purchase-order version updates from recurring LTSA-related entries. Fourth, spare-parts support and stock exposure are analyzed using spare-parts records. Fifth, operational availability and defect information are interpreted with caution because the available data is not symmetrical between the two cases. Finally, Monte Carlo simulation and sensitivity analysis are used to examine uncertainty and robustness in the procurement-related cost exposure mode

7.1 Analytical approach

The thesis uses a quantitative analytical approach because the research question depends on recorded maintenance activity, procurement records, spare-parts information, operational availability indicators, and model-generated outputs. The analysis is based on structured BASSnet CSV records rather than surveys or formal interviews. This makes it possible to compare the two cases using recorded operational and procurement data from the same system environment.

The data cannot be interpreted mechanically. BASSnet is designed for maintenance planning, procurement, and technical follow-up, not primarily for academic research. Job-history records may describe planned maintenance, corrective work, inspections, service activities, cleaning, replacement, or administrative follow-up. Purchase-order rows may represent versions, updates, line items, recurring agreement entries, or ordinary procurement activity. Spare-parts records describe stock and support structure rather than direct consumption. Defect records are only available for one of the two vessel cases.

For this reason, the analysis combines numerical comparison with careful interpretation of data quality and data limitations. The aim is not to prove that one maintenance arrangement is universally better than the other. Instead, the aim is to analyze how the LTSA-based and non-LTSA cases appear under the available BASSnet data, the corrected procurement classification, and the modelling assumptions defined in Chapter 6.

7.2 Structure of the data used in the analysis

Before the two maintenance arrangements can be compared, the data must be reduced to the records most relevant to the thesis question. The analysis is limited to the 2024–2025 period and to the two anonymized vessel cases. The Passenger Vessel represents the LTSA-based case, while the Heavy Lift Vessel represents the non-LTSA case. The comparison is therefore case-based rather than population-based.

The available BASSnet data is summarized in Table 7.1.

Table 7.1: BASSnet data available for analysis

Vessel case	Strategy	Job-history records	Purchase-order rows	Spare-parts records	Defect records
Passenger Vessel	LTSA	2,691	134	104	Not available
Heavy Lift Vessel	Non-LTSA	1,062	83	121	99

Source: Prepared by Evan Elias Håpnes based on the final engine-related BASSnet CSV files.

The table shows that the two vessel cases do not have perfectly symmetrical data availability. The Passenger Vessel has more job-history records and purchase-order rows, while the Heavy Lift Vessel has more spare-parts records and is the only case with defect records. This means that direct defect-frequency comparison is not possible. The Heavy Lift Vessel's defect data can support interpretation of the non-LTSA case, but it cannot be used to claim that the Heavy Lift Vessel had more or fewer defects than the Passenger Vessel.

Job-history records are used as evidence of recorded maintenance activity, not as direct failure counts. Purchase-order records are used to analyze procurement-related cost exposure, but only after correcting for ordinary purchase-order versions and recurring LTSA-related entries. Spare-parts records are treated as stock and support-structure data, not as direct evidence of consumption during 2024–2025.

7.3 Analysis of maintenance activity

Maintenance activity is analyzed through the job-history records. The Passenger Vessel has 2,691 job-history records during the 2024–2025 period, while the Heavy Lift Vessel has 1,062 job-history records. This indicates that the Passenger Vessel has a higher volume of recorded maintenance-related activity in the available BASSnet data.

This difference should be interpreted carefully. A higher number of job-history records does not automatically mean that the Passenger Vessel had worse technical performance. It may also reflect differences in vessel type, engine configuration, planned maintenance structure, operational profile, registration practice, agreement structure, or how maintenance tasks are divided and recorded in BASSnet.

The corrective job-share indicator gives a more focused view of recorded corrective activity. Based on the Job Class field, the Passenger Vessel has 59 corrective job records out of 2,691 total job-history records, giving a corrective share of 2.19%. The Heavy Lift Vessel has 1 corrective job record out of 1,062 total job-history records, giving a corrective share of 0.09%.

This suggests that the Passenger Vessel has more visible corrective maintenance activity in the available job-history data. However, this should still not be interpreted directly as a failure-rate comparison. Corrective job classification depends on how work is categorized in BASSnet, and the two cases may differ in registration practice, maintenance planning, and agreement structure. The result is therefore best interpreted as a recorded corrective activity indicator.

The overdue job-share indicator is also used to analyze maintenance planning pressure. The Passenger Vessel has 658 overdue job records, equal to 24.45% of its job-history records. The Heavy Lift Vessel has 105 overdue job records, equal to 9.89% of its job-history records. This suggests that the Passenger Vessel has a higher visible overdue maintenance share in the available BASSnet job-history data.

However, overdue records should not be interpreted only as poor maintenance performance. They may also reflect differences in planned maintenance volume, job registration detail, operational scheduling, maintenance intervals, system use, or how overdue status is recorded and updated. The overdue indicator is therefore interpreted as a maintenance planning and follow-up indicator, not as a direct measure of technical condition.

7.4 Analysis of procurement-related cost exposure

Procurement-related cost exposure is analyzed using engine-related purchase-order records. This is the main cost-exposure indicator in the model because the final thesis uses BASSnet operational and procurement data. The analysis does not claim to calculate total vessel cost. Instead, it analyzes the procurement-related cost exposure visible in the purchase-order data.

The purchase-order analysis uses a corrected classification. Ordinary purchase-order version updates are treated differently from recurring LTSA-related entries. For ordinary purchase orders, the latest non-cancelled version is used to reduce double-counting. However, recurring LTSA-related monthly, running-hour, and long-lead agreement entries are retained as separate exposure entries when the descriptions and registration pattern indicate that they represent distinct recurring costs rather than duplicate versions of the same order.

This correction is important for the Passenger Vessel. A pure latest-version approach would understate the LTSA-related procurement exposure because recurring monthly or running-hour entries would be treated as duplicate versions. In the corrected model, the Passenger Vessel has USD 1,359,948.51 in LTSA running-hour/monthly exposure, USD 58,332.24 in long-lead/Buyersclub recurring exposure, and USD 755,946.97 in ordinary latest-version non-cancelled purchase-order exposure. This gives a corrected procurement-related cost exposure indicator of USD 2,174,227.72.

The Heavy Lift Vessel has no comparable recurring LTSA-related entries in the available purchase-order data. Its corrected procurement-related cost exposure indicator is therefore based on latest-version non-cancelled purchase orders, giving USD 454,868.63.

The corrected procurement exposure ratio is therefore 4.78. This means that the Passenger Vessel has approximately 4.78 times higher visible procurement-related cost exposure than the Heavy Lift Vessel in the corrected BASSnet purchase-order indicator. This should not be interpreted as a complete vessel-cost ratio. It only reflects the USD-denominated procurement-related cost exposure visible in the available engine-related BASSnet records.

The corrected purchase-order event values also show differences between the cases. The Passenger Vessel has a mean corrected purchase-order event value of USD 18,743.34, while the Heavy Lift Vessel has a mean purchase-order value of USD 6,592.30. The Passenger Vessel therefore has both higher total visible procurement-related cost exposure and higher average purchase-order exposure.

At the same time, the Heavy Lift Vessel has a higher coefficient of variation. The Passenger Vessel has a corrected coefficient of variation of 1.18, while the Heavy Lift Vessel has a coefficient of variation of 1.37. This means that the Heavy Lift Vessel's purchase-order values are relatively more variable compared with its mean, while the Passenger Vessel has the higher total exposure due to the recurring LTSA-related entries and higher ordinary purchase-order exposure.

7.5 Analysis of spare-parts support and stock exposure

Spare-parts support is analyzed through the spare-parts records. These records are not treated as direct consumption history. Instead, they are used to understand stock structure, support burden, and potential spare-parts exposure.

The Passenger Vessel has 104 spare-parts records, while the Heavy Lift Vessel has 121. The difference becomes clearer when looking at records with stock above zero. The

Passenger Vessel has 16 spare-parts records with stock above zero, while the Heavy Lift Vessel has 112. The total stock quantity is also much higher for the Heavy Lift Vessel, with 777 units compared to 30 units for the Passenger Vessel.

The spare-parts stock-value indicator shows the same pattern. The Passenger Vessel has a stock-value indicator of USD 7,804.87, while the Heavy Lift Vessel has a stock-value indicator of USD 244,603.78. This gives a spare-parts stock-value ratio of 31.34, meaning that the Heavy Lift Vessel has a substantially higher visible spare-parts stock-value indicator in the available BASSnet data.

This result should be interpreted carefully. A higher stock-value indicator does not mean that the Heavy Lift Vessel consumed more spare parts during the study period. It means that the available BASSnet spare-parts data shows a larger registered spare-parts stock and support structure. This may reflect vessel type, engine configuration, inventory policy, stock registration practice, agreement structure, or the operational need to hold more parts outside an LTSA structure.

From an LTSA versus non-LTSA perspective, this result is still relevant. A non-LTSA structure may require the operator or technical manager to carry more spare-parts responsibility internally, while an LTSA-based arrangement may shift some support responsibility to the engine manufacturer or service provider. The spare-parts data therefore supports the interpretation that the Heavy Lift Vessel has higher visible internal spare-parts stock exposure in the available BASSnet records.

7.6 Analysis of operational availability and defect data

Operational availability is analyzed as a simplified proxy, not as a full technical availability calculation. The model uses the recorded downtime field in the job-history records. In the final BASSnet CSV files, both vessel cases have zero positive recorded downtime hours in the selected engine-related job-history records.

As shown in Chapter 6, this produces an availability proxy of 100% for both vessel cases. This result should not be interpreted as proof that both vessels had perfect operational availability. Instead, it shows that the available downtime field does not contain usable positive downtime values for the selected records. The availability comparison is therefore inconclusive based on the available BASSnet downtime data.

Defect data is also limited. The Heavy Lift Vessel has 99 defect records, while the Passenger Vessel does not have a comparable defect dataset in the final CSV files. This means that the Heavy Lift Vessel's defect records can be used as supporting evidence for that case, but not as a basis for direct comparison between the two cases. The absence of Passenger Vessel defect records does not mean that no defects occurred. It only means that comparable defect data was not available in the final data foundation.

The availability and defect analysis therefore mainly contributes to the data-quality assessment. It shows that the available BASSnet data supports comparison of maintenance activity, procurement-related cost exposure, and spare-parts structure more strongly than it supports direct comparison of technical availability or defect frequency.

7.7 Monte Carlo and sensitivity-based analysis

Monte Carlo simulation is used to analyze uncertainty in procurement-related cost exposure. The simulation is based on the corrected purchase-order classification from Chapter 6. It does not estimate complete vessel cost. Instead, it simulates possible variation in engine-related procurement exposure based on the observed BASSnet purchase-order values.

The simulation separates recurring agreement-related exposure from ordinary purchase-order exposure. This is necessary because recurring LTSA-related monthly, running-hour, and long-lead agreement entries are not duplicate purchase-order versions. They are included as observed recurring exposure in the Passenger Vessel case, while ordinary purchase orders are simulated as uncertain procurement events.

The Monte Carlo analysis is important because average purchase-order value does not fully describe financial risk. Both vessel cases have high variation in purchase-order values. The Passenger Vessel has a corrected coefficient of variation of 1.18, while the Heavy Lift Vessel has a coefficient of variation of 1.37. This means that both cases have uneven purchase-order distributions, where a limited number of larger orders can strongly affect total procurement exposure.

The top-five purchase-order concentration gives another view of financial risk exposure. Using the corrected purchase-order exposure values, the Passenger Vessel has a top-five concentration of approximately 20.74%, while the Heavy Lift Vessel has a top-five concentration of approximately 35.91%. This indicates that the Heavy Lift Vessel has more concentrated procurement exposure, while the Passenger Vessel has higher total visible procurement-related cost exposure spread across recurring agreement-related entries and ordinary purchase-order records.

Sensitivity analysis is used to test whether the comparison depends heavily on uncertain assumptions. The main sensitivity areas are summarized in Table 7.2.

Table 7.2: Sensitivity areas used in the analysis

Sensitivity area	Reason for testing
Purchase-order exposure break-even factor	Tests how much the Heavy Lift Vessel's purchase-order exposure would need to increase to equal the Passenger Vessel

Spare-parts stock-value weighting	Tests whether the Heavy Lift Vessel's higher spare-parts stock value can overturn the purchase-order exposure difference
Downtime scenario assumptions	Tests how assumed downtime cost could affect the comparison, since recorded downtime values are not usable for full availability analysis
Purchase-order value variation	Tests uncertainty caused by uneven and concentrated purchase-order values

Source: Prepared by Evan Elias Håpnes based on the BASSnet-based model assumptions.

The corrected purchase-order break-even factor is 4.78. This means that the Heavy Lift Vessel's visible purchase-order exposure would need to increase by approximately 378% before it equals the Passenger Vessel's corrected purchase-order exposure.

The spare-parts stock-value break-even factor is 7.26. This means that the Heavy Lift Vessel's higher spare-parts stock-value indicator would need to be weighted very heavily before it overturns the Passenger Vessel's higher corrected purchase-order exposure. Since spare-parts stock is not the same as spare-parts consumption, this result should be interpreted as a robustness test rather than as a direct cost conclusion.

The downtime sensitivity test is also important because the available downtime field contains no positive recorded downtime values for either vessel case. Downtime is therefore treated only as a scenario variable. This allows the thesis to discuss how downtime could affect the comparison without claiming that the available BASSnet downtime field provides a complete technical availability measure.

7.8 Interpretation framework

The results are interpreted using a multi-criteria framework. No single output is treated as sufficient on its own. The two maintenance arrangements are compared across maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, financial risk exposure, and data quality.

A maintenance arrangement with lower visible procurement-related cost exposure may appear financially attractive, but cost exposure alone is not enough. The lower-cost arrangement may require higher internal spare-parts stock, greater procurement responsibility, more supplier coordination, or weaker data visibility. Similarly, a maintenance arrangement with higher visible procurement-related cost exposure may still provide value if it improves predictability, technical support, spare-parts access, risk allocation, or access to manufacturer expertise.

The interpretation framework therefore does not ask only which arrangement is cheapest. It asks which arrangement appears more favorable under the available BASSnet data, while also considering uncertainty, availability limitations, spare-parts support, procurement concentration, and data quality. This is consistent with the purpose of the thesis as a decision-support study rather than a purely accounting-based comparison.

The analysis also follows the supervisor feedback by avoiding client-specific vessel names and by treating the cases as anonymized vessel types within the Wilhelmsen Ship Management portfolio. The Passenger Vessel and the Heavy Lift Vessel are not assumed to be directly comparable in every respect. The results must therefore be interpreted as a structured case-based comparison, not as a universal conclusion about all LTSA-based and non-LTSA maintenance arrangements.

8.0 Results

This chapter presents the empirical and model-based results from the comparison between the LTSA-based and non-LTSA maintenance arrangements. The results are structured around the main dimensions of the thesis: maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, financial risk exposure, and data quality. The Passenger Vessel represents the LTSA-based case, while the Heavy Lift Vessel represents the non-LTSA case.

The results are based only on the final engine-related BASSnet CSV files for the 2024–2025 period. The datasets include job-history records, purchase-order records, spare-parts records, and defect records where available. All monetary values are presented in USD. The results should be interpreted as a case-based comparison of available operational and procurement records, not as a complete financial or engineering evaluation of the two vessels.

8.1 Empirical data overview

The empirical data is not fully symmetrical between the two vessel cases. Both vessels have job-history records, purchase-order records, and spare-parts records. However, only the Heavy Lift Vessel has defect records in the final BASSnet CSV data. This means that defect-level results can support the Heavy Lift Vessel case, but they cannot be used for a direct defect-frequency comparison between the two vessels.

Table 8.1: BASSnet data available for the results

Vessel case	Strategy	Job-history records	Purchase-order rows	Spare-parts records	Defect records
Passenger Vessel	LTSA	2,691	134	104	Not available
Heavy Lift Vessel	Non-LTSA	1,062	83	121	99

Source: Prepared by Evan Elias Håpnes based on the final engine-related BASSnet CSV files

Figure 8.1 BASSnet record-count overview

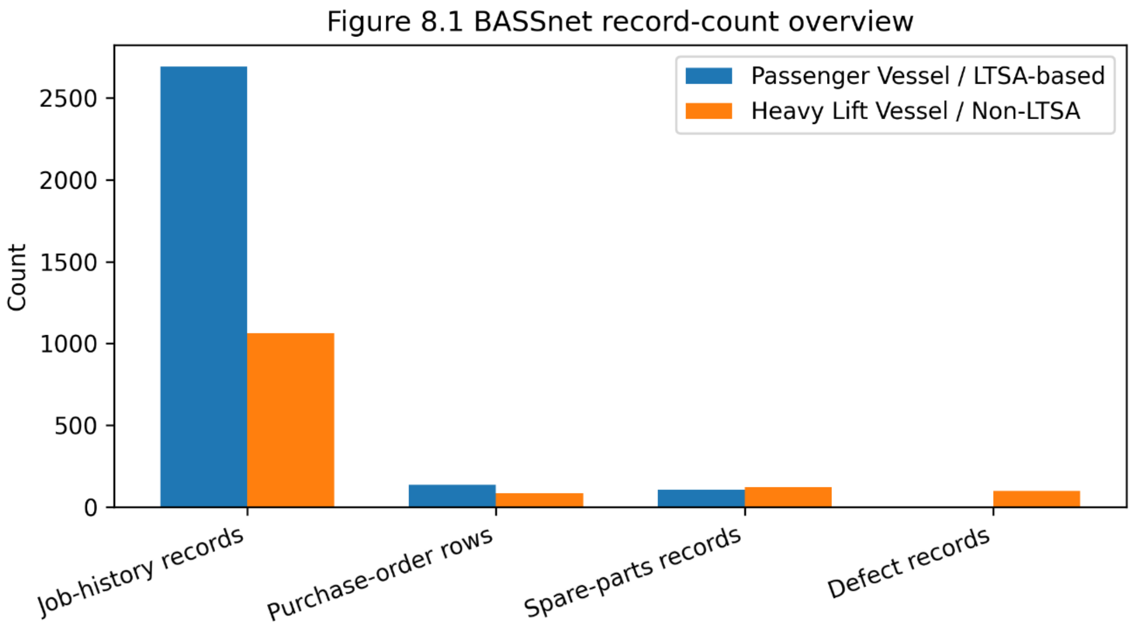

Figure 8.1 shows the number of BASSnet records available for each vessel case and dataset category. The Passenger Vessel has more job-history records, while the Heavy Lift Vessel has more spare-parts records and is the only case with defect records. The Passenger Vessel defect value is shown as not available, not as zero defects. This supports the interpretation that the two cases have unequal data availability and should be compared as case-based evidence rather than as a fully symmetrical statistical dataset.

The Passenger Vessel has the larger job-history dataset, with 2,691 records compared to 1,062 for the Heavy Lift Vessel. This does not mean that the Passenger Vessel had more failures. Job-history records can include planned maintenance, inspections, cleaning, service tasks, replacements, corrective work, and administrative follow-up. The records are therefore interpreted as evidence of recorded maintenance activity, not as direct failure counts.

The Heavy Lift Vessel has more spare-parts records and is the only case with defect records. The absence of Passenger Vessel defect records should not be interpreted as evidence that no defects occurred. It only means that comparable defect-level data was not available in the final CSV files.

8.2 Maintenance activity results

Maintenance activity was analyzed using job-history records. The Passenger Vessel has 2,691 job-history records across the 2024–2025 period, while the Heavy Lift Vessel has 1,062 job-history records.

Table 8.2: Maintenance activity indicators

Indicator	Passenger Vessel, LTSA	Heavy Lift Vessel, non-LTSA
Job-history records	2,691	1,062
Corrective job records	59	1
Corrective job share	2.19%	0.09%
Overdue job records	658	105
Overdue job share	24.45%	9.89%
Recorded downtime hours	0	0

Source: Prepared by Evan Elias Håpnes based on the final BASSnet job-history CSV files.

The Passenger Vessel has a higher number of recorded job-history records and a higher overdue job share. The corrective job share is also higher for the Passenger Vessel, at 2.19%, compared to 0.09% for the Heavy Lift Vessel.

These results should be interpreted carefully. A higher number of job-history records may reflect more recorded maintenance activity, but it may also reflect differences in vessel type, engine configuration, maintenance planning structure, operational profile, registration practice, agreement structure, or how tasks are divided in BASSnet. Similarly, overdue records should be interpreted as a maintenance planning and follow-up indicator, not as a direct measure of technical condition.

8.3 Procurement-related cost exposure results

Procurement-related cost exposure was analyzed using engine-related purchase-order records. The purchase-order data was corrected by separating ordinary purchase-order version updates from recurring LTSA-related agreement entries. This correction is important because a pure latest-version-only approach would understate the Passenger Vessel's visible LTSA-related procurement exposure.

For ordinary purchase orders, the latest non-cancelled version is used to reduce double-counting. However, recurring LTSA-related monthly, running-hour, and long-lead agreement entries are retained as separate exposure entries when the descriptions and registration pattern indicate that they represent distinct recurring costs rather than duplicate versions of the same order.

Table 8.3: Purchase-order cost exposure indicators

Indicator	**Passenger Vessel, LTSA**	**Heavy Lift Vessel, non-LTSA**
Purchase-order rows	134	83
Ordinary latest-version, non-cancelled purchase orders	50	69
Recurring agreement-related entries retained separately	66	0
LTSA running-hour/monthly exposure, USD	1,359,948.51	Not applicable
Long-lead/Buyersclub recurring exposure, USD	58,332.24	Not applicable
Ordinary purchase-order exposure, USD	755,946.97	454,868.63
Corrected purchase-order exposure indicator, USD	2,174,227.72	454,868.63
Mean corrected purchase-order event value, USD	18,743.34	6,592.30
Standard deviation of corrected purchase-order event value, USD	22,204.15	9,045.92
Coefficient of variation	1.18	1.37
Top-five purchase-order concentration	20.74%	35.91%

Source: Prepared by Evan Elias Håpnes based on the final BASSnet purchase-order CSV files.

The corrected purchase-order exposure indicator for the Passenger Vessel is USD 2,174,227.72. This consists of USD 1,359,948.51 in LTSA running-hour/monthly exposure, USD 58,332.24 in long-lead/Buyers club recuring exposure, and USD 755,946.97 in ordinary latest-version non-cancelled purchase-order exposure. The Heavy Lift Vessel has no comparable recurring LTSA-related entries in the available purchase-order data, so its corrected purchase-order exposure indicator remains USD 454,868.63.

The corrected procurement exposure ratio is:

$$PER = \frac{2{,}174{,}227.72}{454{,}868.63} = 4.78$$

This means that the Passenger Vessel has approximately 4.78 times higher visible procurement-related cost exposure than the Heavy Lift Vessel in the corrected BASSnet purchase-order indicator. This should not be interpreted as a complete vessel-cost ratio. It only reflects the USD-denominated procurement-related cost exposure visible in the available engine-related BASSnet records.

Figure 8.2 Procurement-related cost exposure indicator

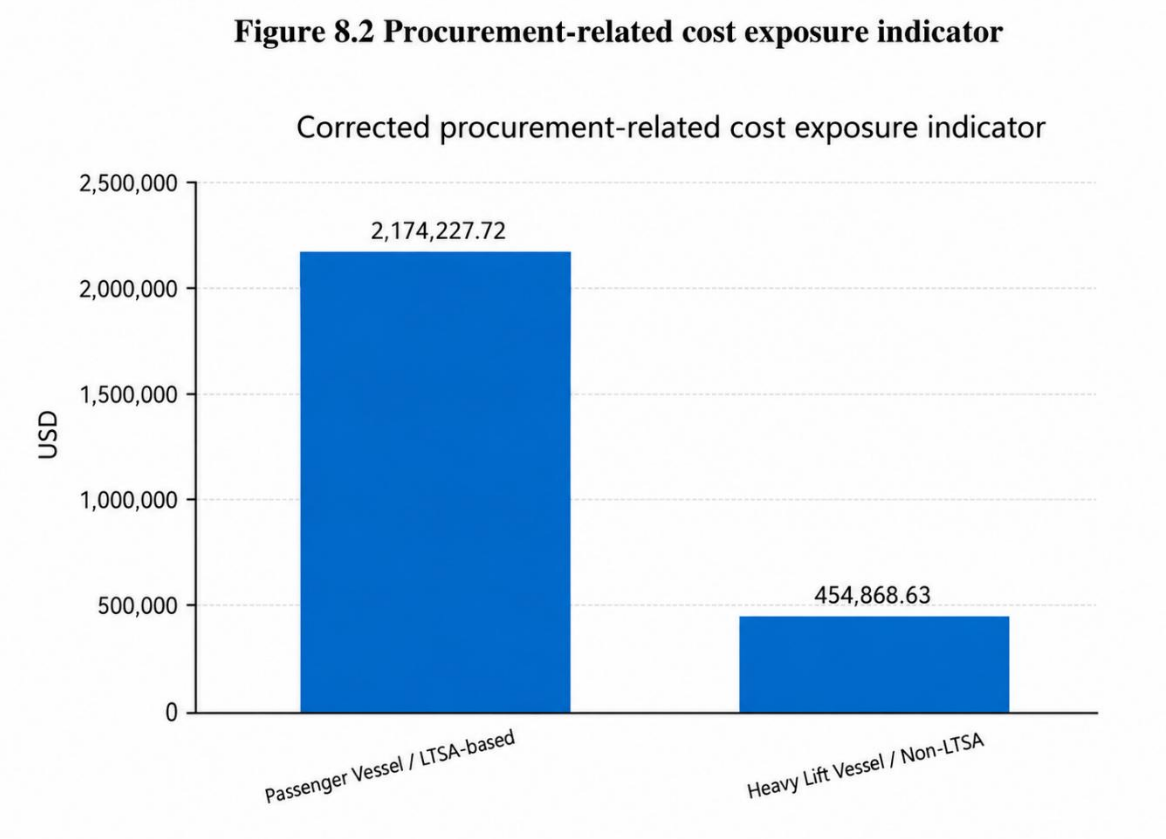

Figure 8.2 compares the corrected procurement-related cost exposure indicator for the two vessel cases. The Passenger Vessel has a corrected visible procurement-related cost exposure of USD 2,174,227.72, while the Heavy Lift Vessel has USD 454,868.63. This shows that the Passenger Vessel has substantially higher visible procurement-related cost exposure in the corrected purchase-order dataset.

The mean corrected purchase-order event value is also higher for the Passenger Vessel. The Passenger Vessel has a mean corrected purchase-order event value of USD 18,743.34, while the Heavy Lift Vessel has a mean purchase-order value of USD 6,592.30. However, the coefficient of variation is higher for the Heavy Lift Vessel, at 1.37 compared to 1.18 for the Passenger Vessel. This means that the Heavy Lift Vessel has relatively more variation in purchase-order values compared with its mean, while the Passenger Vessel has the higher total visible procurement-related cost exposure.

The top-five concentration also differs between the cases. The Passenger Vessel has a corrected top-five concentration of 20.74%, while the Heavy Lift Vessel has a top-five concentration of 35.91%. This suggests that the Heavy Lift Vessel's procurement exposure is more concentrated in a smaller number of high-value purchase orders, while the Passenger Vessel's higher total exposure is spread across recurring agreement-related entries and ordinary purchase-order records.

8.4 Spare-parts support and stock exposure results

Spare-parts records were analyzed as stock and support-structure data. They are not interpreted as direct consumption history. The purpose is to understand the visible spare-parts support burden in the available BASSnet records.

Table 8.4: Spare-parts support indicators

Indicator	Passenger Vessel, LTSA	Heavy Lift Vessel, non-LTSA
Spare-parts records	104	121
Spare-parts records with stock above zero	16	112
Total stock quantity	30	777
Spare-parts stock-value indicator, USD	7,804.87	244,603.78

Source: Prepared by Evan Elias Håpnes based on the final BASSnet spare-parts CSV files.

Figure 8.3 Spare-parts stock value indicator

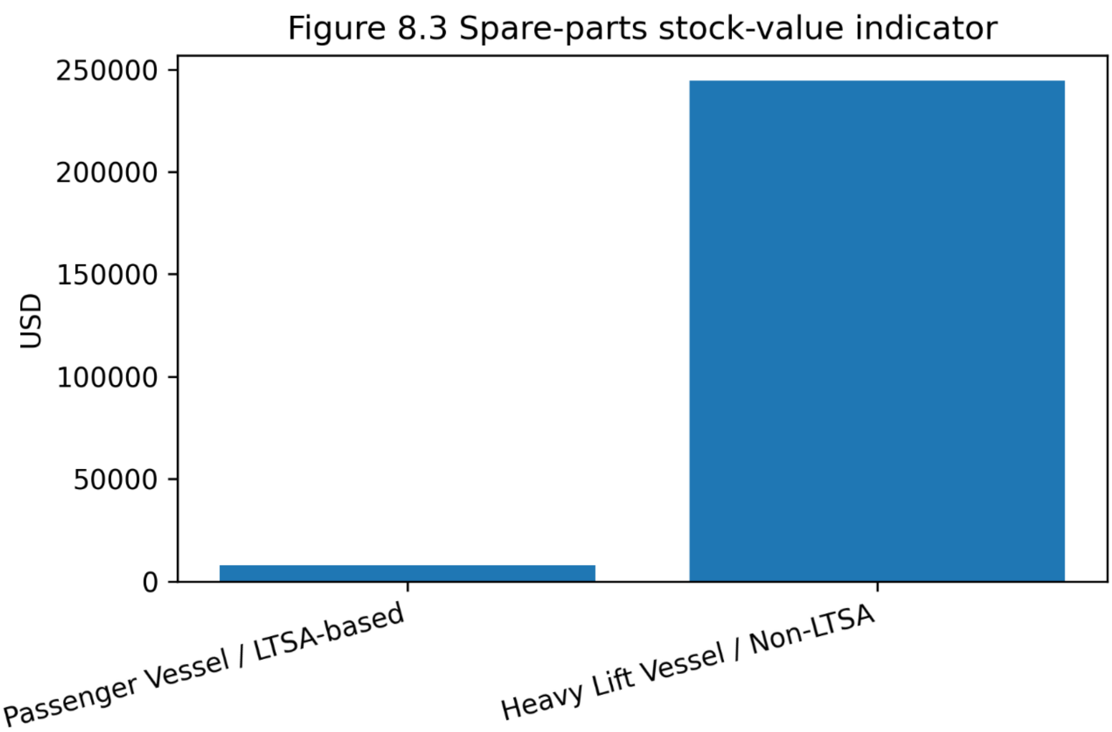

Figure 8.3 compares the spare-parts stock-value indicator for the two vessel cases. The Heavy Lift Vessel has a much higher visible spare-parts stock-value indicator than the Passenger Vessel. This suggests that the non-LTSA case carries a larger visible spare-parts stock and support-structure exposure in the available BASSnet data. The result should not be interpreted as spare-parts consumption during the period, because spare-parts records are treated as stock and support-structure data.

The Heavy Lift Vessel has a much larger visible spare-parts stock position than the Passenger Vessel. It has 112 spare-parts records with stock above zero, compared to 16 for the Passenger Vessel. The total stock quantity is also much higher for the Heavy Lift Vessel, with 777 units compared to 30 units for the Passenger Vessel.

The spare-parts stock-value ratio is:

$$SPR = \frac{244603.78}{7804.87} = 31.34$$

This means that the Heavy Lift Vessel has a spare-parts stock-value indicator approximately 31.34 times higher than the Passenger Vessel in the available BASSnet data.

This does not mean that the Heavy Lift Vessel consumed more spare parts during the period. It means that the visible stock and support-structure exposure is higher. This may reflect vessel type, engine configuration, inventory policy, operational requirements, agreement structure, or how spare parts are registered in BASSnet.

In the non-LTSA case, this higher spare-parts stock-value indicator can also be interpreted as a risk exposure. Since the Heavy Lift Vessel operates outside an LTSA structure, a larger share of spare-parts planning, procurement timing, supplier coordination, and cost uncertainty may remain with the operator or technical manager. The higher visible stock value may therefore suggest that the non-LTSA arrangement requires more internal inventory preparedness to reduce the risk of unavailable critical parts. If critical components are not available when needed, the vessel may face urgent purchasing, higher procurement prices, longer repair times, or increased operational disruption.

At the same time, holding a larger spare-parts stock also creates financial risk. Expensive spare parts tie up capital and may become slow-moving, overstocked, or less relevant if maintenance needs change. This means that the non-LTSA case may provide more flexibility and direct control, but it also places more responsibility on the company to balance stockout risk against the cost of holding high-value inventory. In this way, the higher spare-parts stock-value indicator suggests that the non-LTSA arrangement may carry greater self-managed inventory and spare-parts risk, especially because the company must balance parts-availability risk against the financial cost of holding high-value spare parts.

8.5 Defects and availability results

Defect records are available only for the Heavy Lift Vessel. The final BASSnet defect file contains 99 defect records.

Table 8.5: Heavy Lift Vessel defect record overview

Defect indicator	Heavy Lift Vessel
Total defect records	99
Closed defect records	79
Completed defect records	10
New defect records	10
Overdue defect records	5
Critical defect records	24
Stand-by critical defect records	13
Non-critical defect records	54
Records without priority classification	8

Source: Prepared by Evan Elias Håpnes based on the final BASSnet defect CSV file.

The defect data provides useful supporting evidence for the Heavy Lift Vessel case. However, it cannot be used for direct comparison because the Passenger Vessel does not have a comparable defect dataset in the final CSV files. The absence of comparable Passenger Vessel defect records does not mean that no defects occurred. It only means that defect-level records were not available for both cases.

Operational availability was analyzed using the downtime field in the job-history records. Both vessel cases have zero positive recorded downtime hours in the selected engine-related BASSnet job-history data.

Since the recorded downtime value is zero for both cases, the availability proxy gives the same result for both:

$$A_{Passenger} = A_{HeavyLift} = \frac{17544 - 0}{17544} = 100\%$$

Table 8.6: Availability proxy result

Indicator	Passenger Vessel, LTSA	Heavy Lift Vessel, non-LTSA
Operating horizon, hours	17,544	17,544
Positive recorded downtime hours	0	0
Availability proxy	100.00%	100.00%

Source: Prepared by Evan Elias Håpnes based on the final BASSnet job-history CSV files.

This result should not be interpreted as proof that both vessels had perfect operational availability. It only shows that the selected BASSnet downtime field does not contain usable positive downtime values for the engine-related records. The availability result is therefore mainly a data-quality finding, and the direct availability comparison should be treated as inconclusive.

8.6 Monte Carlo procurement-exposure results

Monte Carlo simulation was used to analyze uncertainty in procurement-related cost exposure. The simulation is based on the corrected purchase-order classification from Chapter 6. It does not estimate complete vessel cost. Instead, it simulates possible variation in visible engine-related procurement exposure based on the observed BASSnet purchase-order values.

The corrected simulation separates recurring agreement-related exposure from ordinary purchase-order exposure. Recurring LTSA-related monthly, running-hour, and long-lead agreement entries are included as observed recurring exposure for the Passenger Vessel. Ordinary purchase orders are simulated as uncertain procurement events. This correction is necessary because recurring LTSA-related entries are not duplicate purchase-order versions.

The simulation used 10,000 iterations with a fixed random seed for reproducibility. The number of ordinary purchase-order events was modelled using a Poisson distribution, and purchase-order values were sampled from the observed ordinary purchase-order value distribution for each vessel case.

Table 8.7: Monte Carlo procurement-exposure summary

Monte Carlo output	**Passenger Vessel, LTSA**	**Heavy Lift Vessel, non-LTSA**
Mean simulated exposure, USD	2,169,870.79	454,327.73
Median simulated exposure, USD	2,151,475.36	449,822.85
Standard deviation, USD	207,780.37	92,630.63
5th percentile, USD	1,857,507.98	310,961.59
95th percentile, USD	2,537,313.49	616,492.15

Source: Prepared by Evan Elias Håpnes based on the BASSnet-based Monte Carlo model.

Figure 8.4 Monte Carlo mean and 5th-95th percentile range

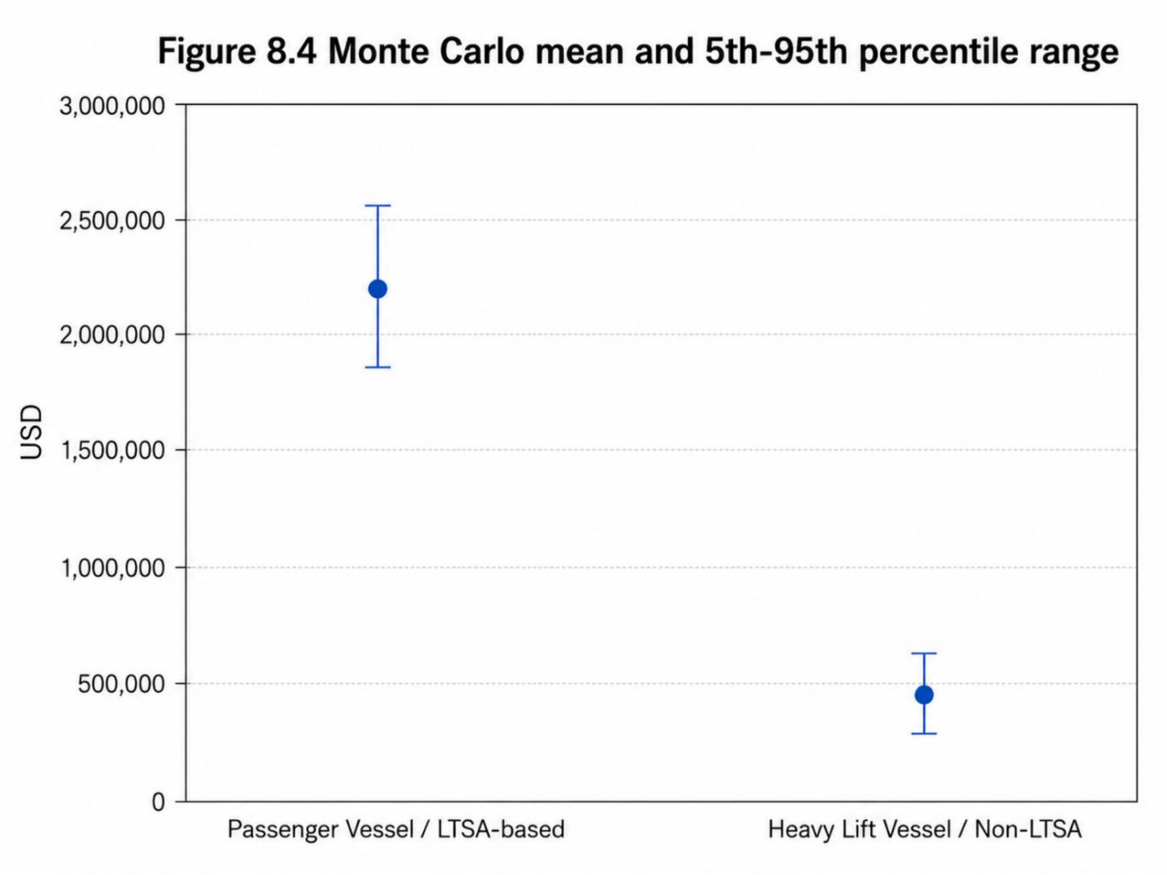

Figure 8.4 shows the corrected Monte Carlo mean simulated procurement exposure and the central 5th–95th percentile range for each vessel case. The Passenger Vessel has both a higher mean simulated exposure and a wider absolute uncertainty range than the Heavy Lift Vessel. This supports the interpretation that the Passenger Vessel has higher visible procurement-related cost exposure under the corrected model assumptions.

The Monte Carlo results show that the Passenger Vessel has a mean simulated procurement-related cost exposure of USD 2,169,870.79, while the Heavy Lift Vessel has a mean simulated exposure of USD 454,327.73. The Passenger Vessel also has a wider absolute spread of possible outcomes, with a standard deviation of USD 207,780.37 compared to USD 92,630.63 for the Heavy Lift Vessel. The paired simulation difference is summarized in Table 8.8.

Table 8.8: Paired Monte Carlo difference summary

Metric	Result
Mean Passenger Vessel minus Heavy Lift Vessel exposure, USD	1,715,543.07
5th percentile difference, USD	1,371,638.80
95th percentile difference, USD	2,109,030.81
Probability that Passenger Vessel exposure is higher	100.00%

Probability that Heavy Lift Vessel exposure is higher	0.00%

Source: Prepared by Evan Elias Håpnes based on the BASSnet-based Monte Carlo model.

The paired simulation shows that the Passenger Vessel has higher simulated procurement-related cost exposure in all simulation runs. This supports the result from the deterministic purchase-order analysis. However, the result should still be interpreted as procurement-related cost exposure only, not as a complete vessel-cost conclusion.

8.7 Sensitivity analysis results

Sensitivity analysis was used to test how robust the comparison is when important assumptions are changed. The sensitivity analysis focuses on purchase-order exposure, spare-parts stock-value exposure, and downtime assumptions.

Table 8.9: Sensitivity analysis summary

Sensitivity test	Result	Interpretation
Heavy Lift Vessel purchase-order break-even factor	4.78	Heavy Lift Vessel purchase-order exposure would need to increase by approximately 378% to equal the Passenger Vessel's corrected purchase-order exposure
Passenger Vessel purchase-order reduction factor	0.21	Passenger Vessel purchase-order exposure would need to fall to about 20.9% of its corrected observed value to equal the Heavy Lift Vessel
Spare-parts stock-value break-even factor	7.26	Heavy Lift Vessel spare-parts stock value must be weighted very heavily before it overturns the corrected purchase-order exposure difference
Recorded downtime sensitivity	Inconclusive	The available downtime field contains zero positive recorded downtime hours for both cases

Source: Prepared by Evan Elias Håpnes based on the BASSnet-based model assumption

Figure 8.5 Sensitivity break even indicato

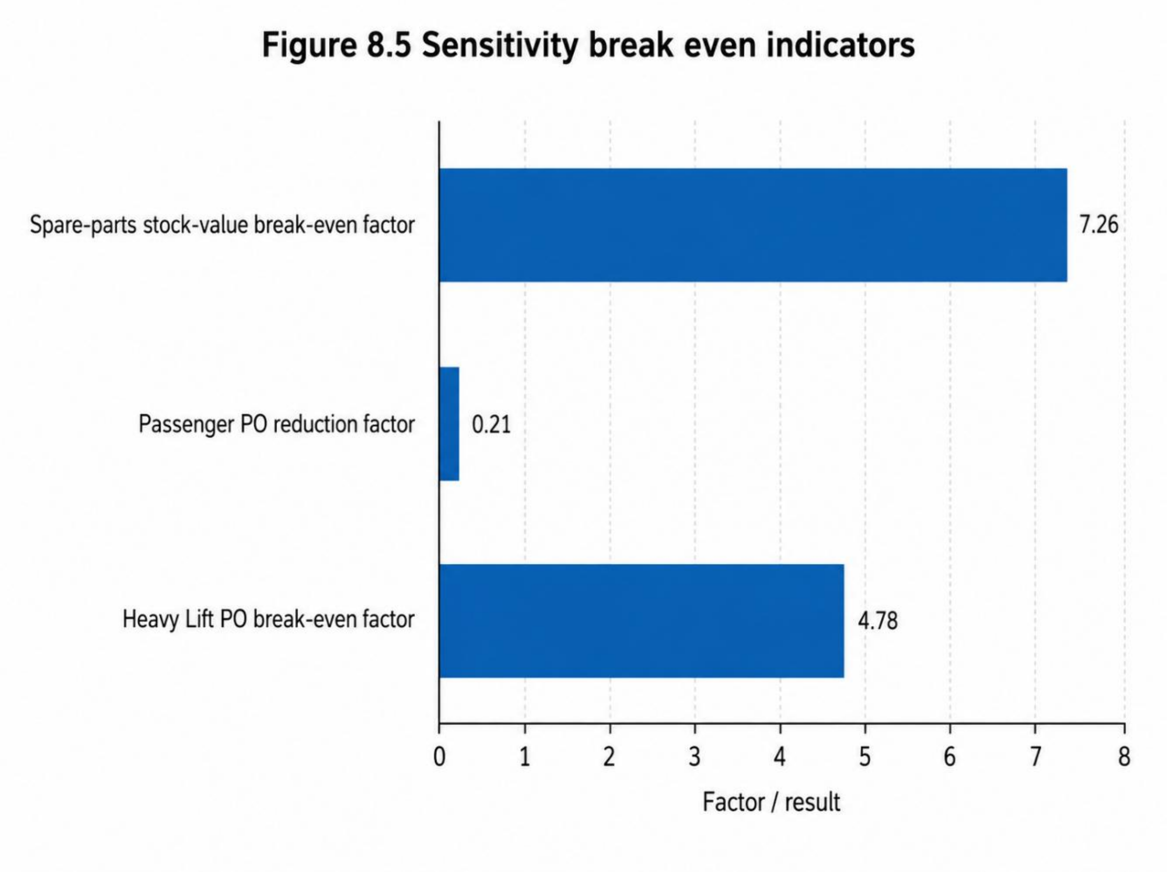

Figure 8.5 summarizes the corrected sensitivity break-even indicators. The Heavy Lift Vessel purchase-order exposure would need to increase by a factor of 4.78 to equal the Passenger Vessel's corrected purchase-order exposure. The Passenger Vessel purchase-order exposure would need to fall to 0.21 of its corrected observed value to equal the Heavy Lift Vessel. The spare-parts stock-value break-even factor is 7.26, meaning that spare-parts stock exposure must be weighted very heavily before it overturns the corrected purchase-order exposure difference.

The corrected purchase-order break-even factor is 4.78. This means that the Heavy Lift Vessel's visible purchase-order exposure would need to increase by approximately 378% before it equals the Passenger Vessel's corrected purchase-order exposure.

The spare-parts stock-value break-even factor is 7.26. This means that the Heavy Lift Vessel's higher spare-parts stock-value indicator would need to be weighted very heavily before it overturns the Passenger Vessel's higher corrected purchase-order exposure. Since spare-parts stock is not the same as spare-parts consumption, this result should be interpreted as a robustness test rather than as a direct cost conclusion.

The downtime sensitivity result is limited by the available data. Since both cases have zero positive recorded downtime hours in the selected job-history records, the downtime field cannot support a reliable direct availability comparison. Downtime can still be discussed as a scenario-based risk, but not as an observed difference between the two cases.

8.8 Summary of main results

The main results are summarized in Table 8.10.

Table 8.10: Summary of main results

Result area	Passenger Vessel, LTSA	Heavy Lift Vessel, non-LTSA	Main result
Job-history records	2,691	1,062	Passenger Vessel has higher recorded maintenance activity
Corrective job share	2.19%	0.09%	Passenger Vessel has higher corrective share in the available job-history data
Overdue job share	24.45%	9.89%	Passenger Vessel has higher overdue maintenance share
Corrected purchase-order exposure, USD	2,174,227.72	454,868.63	Passenger Vessel has higher visible procurement-related cost exposure
Mean corrected purchase-order event value, USD	18,743.34	6,592.30	Passenger Vessel has higher average corrected purchase-order event value
Coefficient of variation	1.18	1.37	Heavy Lift Vessel has higher relative purchase-order variation
Top-five PO concentration	20.74%	35.91%	Heavy Lift Vessel has more concentrated purchase-order exposure
Spare-parts records	104	121	Heavy Lift Vessel has more spare-parts records
Spare-parts stock-value indicator, USD	7,804.87	244,603.78	Heavy Lift Vessel has much higher visible stock exposure
Defect records	Not available	99	Defect comparison is not possible
Availability proxy	100.00%	100.00%	Availability comparison is inconclusive due to zero positive recorded downtime
Mean Monte Carlo exposure, USD	2,169,870.79	454,327.73	Passenger Vessel has higher simulated procurement exposure

Probability Passenger Vessel exposure is higher	100.00%	—	Monte Carlo supports higher Passenger Vessel procurement exposure
Corrected PO break-even factor	—	4.78	Heavy Lift exposure would need to increase by about 378% to equal Passenger corrected exposure
Spare-parts stock-value break-even factor	—	7.26	Spare-parts stock value must be weighted heavily to overturn corrected PO exposure difference

Overall, the results show that the Passenger Vessel has higher visible procurement-related cost exposure and higher recorded maintenance activity in the available BASSnet data. The Heavy Lift Vessel has higher visible spare-parts stock exposure and is the only case with defect records.

The results do not prove that one maintenance arrangement is universally better than the other. The vessels differ in type, engine configuration, operational profile, agreement structure, and data availability. The findings should therefore be interpreted as a structured case-based comparison of available BASSnet records. The strongest results concern maintenance activity, corrected purchase-order exposure, spare-parts stock exposure, and procurement-related uncertainty. The weakest result area is operational availability, because the available downtime field does not provide usable positive downtime observations for either vessel case. The availability result should therefore be treated as inconclusive.

9.0 Discussion

This chapter discusses the results presented in Chapter 8 in relation to the research question, theoretical framework, literature review, and methodological choices made in the thesis. While Chapter 8 presented the empirical and model-based results, this chapter interprets what those results mean and how reliable they are for maintenance decision-making.

The discussion is based on the comparison between the Passenger Vessel, representing the LTSA-based case, and the Heavy Lift Vessel, representing the non-LTSA case. The results are discussed across the main dimensions of the thesis: maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, financial risk exposure, and data quality. Particular attention is given to the limitations of the available BASSnet data, especially the lack of comparable defect data for the Passenger Vessel and the limited usefulness of the recorded downtime field.

9.1 Discussion of the main findings

The main finding is that the Passenger Vessel has higher visible procurement-related cost exposure in the available BASSnet purchase-order data, while the Heavy Lift Vessel has higher visible spare-parts stock exposure. After correcting the purchase-order classification, the Passenger Vessel has a corrected purchase-order exposure indicator of USD 2,174,227.72, compared with USD 454,868.63 for the Heavy Lift Vessel. This gives a corrected procurement exposure ratio of 4.78, meaning that the Passenger Vessel has approximately 4.78 times higher visible procurement-related cost exposure than the Heavy Lift Vessel in the available BASSnet records.

This finding differs from the earlier latest-version-only interpretation because the corrected model separates ordinary purchase-order version updates from recurring LTSA-related agreement entries. This distinction is important. A pure latest-version approach would understate the Passenger Vessel's LTSA-related exposure because recurring monthly, running-hour, and long-lead agreement entries may represent distinct recurring costs rather than duplicate versions of the same purchase order. The corrected result therefore provides a more realistic view of the visible procurement-related exposure connected to the LTSA-based case.

The Monte Carlo simulation supports the same overall pattern. The mean simulated procurement-related cost exposure is USD 2,169,870.79 for the Passenger Vessel and USD 454,327.73 for the Heavy Lift Vessel. The paired simulation shows that the Passenger Vessel has higher simulated procurement-related exposure in 100.00% of the simulation runs. This supports the deterministic purchase-order result and suggests that the corrected procurement exposure difference is robust within the assumptions of the model.

However, this finding should not be interpreted as proof that LTSA-based arrangements are generally more expensive than non-LTSA arrangements. The result applies only to the selected vessel cases, the 2024–2025 period, the available BASSnet records, and the modelling assumptions used in this thesis. The analysis does not include every cost element connected to engine maintenance, and it does not represent a complete vessel-level cost calculation. It shows visible procurement-related cost exposure in the available engine-related BASSnet records.

The spare-parts result gives a different perspective. The Heavy Lift Vessel has a spare-parts stock-value indicator of USD 244,603.78, compared with USD 7,804.87 for the Passenger Vessel. This gives a stock-value ratio of 31.34. This does not mean that the Heavy Lift Vessel consumed more spare parts during the study period. It means that the visible spare-parts stock and support structure is much larger in the available BASSnet data. This may reflect a greater need for internal spare-parts responsibility under a non-LTSA arrangement.

The maintenance activity results also show important differences. The Passenger Vessel has 2,691 job-history records, compared with 1,062 for the Heavy Lift Vessel. The Passenger Vessel also has a higher overdue job share, at 24.45%, compared with 9.89% for the Heavy Lift Vessel. At first, this may appear to indicate more maintenance pressure for the Passenger Vessel. However, these findings should not be interpreted directly as evidence of more failures. A higher number of job-history records may also reflect a more detailed planned maintenance structure, more systematic registration in BASSnet, different engine configuration, or stronger follow-up requirements under the LTSA arrangement.

The reasoning behind this interpretation is that maintenance records show how work is registered and managed, not only how often technical problems occur. In an LTSA-based structure, maintenance activities may be more formally planned, documented, and followed up through defined service routines. This can increase the number of visible records without necessarily meaning that the vessel performs worse technically. The higher overdue share may therefore indicate a larger administrative and planning workload, or stricter follow-up of planned maintenance tasks, rather than direct evidence of lower reliability.

For the non-LTSA case, the lower number of job-history records and lower overdue share should also be interpreted carefully. It may suggest fewer visible maintenance activities in the available BASSnet data, but it does not automatically prove that the Heavy Lift Vessel had fewer technical issues or lower maintenance demand. Some activities may be registered differently, handled through other routines, or affected by differences in vessel operation and data availability. The main implication is therefore that maintenance activity must be evaluated together with procurement exposure, spare-parts responsibility, defect records, and data quality. This supports the thesis argument that LTSA and non-LTSA arrangements cannot be judged from record counts alone.

9.2 Discussion in relation to the research question

The research question asks how an LTSA-based maintenance arrangement compares with a non-LTSA maintenance arrangement for MAN marine engines through available BASSnet records, in terms of maintenance activity, procurement-related cost exposure, spare-parts support, operational availability indicators, and financial risk exposure. Based on the current analysis, the answer must be conditional and case-specific.

For procurement-related cost exposure, the Passenger Vessel shows substantially higher visible exposure than the Heavy Lift Vessel. This is shown both by the corrected deterministic purchase-order indicator and by the Monte Carlo simulation. The corrected purchase-order exposure indicator is USD 2,174,227.72 for the Passenger Vessel and USD 454,868.63 for the Heavy Lift Vessel. This means that the LTSA-based case has higher visible procurement-related exposure in the available BASSnet purchase-order records.

For spare-parts support, the result points in the opposite direction. The Heavy Lift Vessel has a much higher spare-parts stock-value indicator, more spare-parts records with stock above zero, and a higher total stock quantity. This suggests that the non-LTSA case may require more visible internal spare-parts support and inventory responsibility. This is important because lower visible purchase-order exposure does not necessarily mean lower total maintenance burden. A non-LTSA arrangement may shift more responsibility to internal planning, procurement, stockholding, supplier follow-up, and refurbishment coordination.

For operational availability, the result is less conclusive. The available BASSnet job-history data shows zero positive recorded downtime hours for both vessel cases. This produces an availability proxy of 100% for both vessels, but this should not be interpreted as proof of perfect operational availability. It only shows that the selected downtime field does not contain usable positive downtime values. Therefore, the thesis cannot make a strong conclusion about which arrangement provides better real operational availability.

For financial risk exposure, the results show two different types of exposure. The Passenger Vessel has higher absolute procurement-related cost exposure and higher simulated procurement exposure. At the same time, the Heavy Lift Vessel has higher relative purchase-order variation, higher top-five purchase-order concentration, and much higher spare-parts stock exposure. This means that financial risk cannot be judged only by purchase-order totals. It must also consider cost concentration, uncertainty, supplier dependence, stock responsibility, and data visibility.

Overall, the research question can be answered as follows: within the available BASSnet data and the corrected modelling framework, the LTSA-based case shows higher visible procurement-related cost exposure and higher recorded maintenance activity, while the non-LTSA case shows higher visible spare-parts stock exposure and more concentrated purchase-order values. The available results do not prove that one arrangement is universally better than the other, but they show that the two arrangements create different types of maintenance and financial exposure.

9.3 Discussion in relation to theory and literature

The results support the theoretical argument that maintenance arrangements should be evaluated as asset-management decisions rather than as narrow cost comparisons. A simple comparison of one cost category would not fully capture the differences between the two cases. The results show that maintenance arrangements affect recorded maintenance activity, procurement exposure, spare-parts support, data visibility, risk allocation, and uncertainty.

From an engineering asset management perspective, the findings show that maintenance decisions involve trade-offs between cost exposure, operational support, risk allocation, and technical responsibility. The LTSA-based case has higher visible procurement-related exposure, but this may reflect structured agreement payments, manufacturer involvement, technical support, and recurring service-related obligations. The non-LTSA case has lower visible purchase-order exposure, but higher visible spare-parts stock exposure and potentially greater internal responsibility for planning, procurement, and support.

From a reliability, availability, maintainability, and dependability perspective, the results show why availability should be treated carefully. The thesis includes an availability proxy because operational availability is central to the research question. However, the available BASSnet data does not provide a complete technical availability measure. This supports the theoretical point that availability is more complex than recorded downtime fields alone. Real availability depends on technical condition, repair duration, spare-parts availability, operational schedules, and how downtime is recorded.

The findings are also relevant to service contract theory. LTSA-based arrangements are often expected to provide structure, predictability, and technical support from the engine manufacturer or service provider. The results show that this does not automatically mean lower visible procurement exposure in the available records. The Passenger Vessel still has substantial procurement-related exposure after correcting for recurring LTSA-related entries. This suggests that an LTSA arrangement may change how responsibility and risk are structured, but it does not necessarily remove all procurement or spare-parts-related activity from the operator or technical manager.

The spare-parts findings support the importance of maritime spare-parts logistics. The Heavy Lift Vessel's much higher spare-parts stock-value indicator suggests that spare-parts support remains a key issue under non-LTSA arrangements. This fits the theory that spare-parts availability is central to maintenance effectiveness, especially when vessels operate as moving assets and supply chains are geographically dispersed. The comparison therefore shows that maintenance arrangements cannot be evaluated without considering spare-parts support, procurement structure, and inventory responsibility.

At the same time, this thesis differs from much of the general literature because it is based on a specific company case and real operational data. The contribution of the thesis is not to prove a universal relationship between LTSA-based and non-LTSA arrangements.

Instead, it shows how these arrangements can be compared using available BASSnet records, corrected purchase-order classification, Monte Carlo simulation, and sensitivity analysis under real data limitations.

9.4 Expected and unexpected findings

Some findings were expected. It was expected that the LTSA-based and non-LTSA cases would differ in maintenance structure because the two arrangements allocate responsibility differently. An LTSA-based arrangement may provide more structured support through the engine manufacturer or service provider, while a non-LTSA arrangement may require more internal responsibility for spare-parts planning, procurement, and refurbishment coordination.

It was also expected that the non-LTSA case might show higher spare-parts responsibility. This appears in the results through the Heavy Lift Vessel's higher spare-parts stock-value indicator. The Heavy Lift Vessel has more spare-parts records with stock above zero and a much higher total stock-value indicator than the Passenger Vessel. This supports the expectation that non-LTSA arrangements may require more visible internal stock support.

The more surprising finding is that the Passenger Vessel has much higher visible procurement-related exposure after correcting the purchase-order classification. One possible explanation is that the LTSA does not cover all maintenance, repair, and spare-parts activity. Another explanation is that the Passenger Vessel has a different operational profile, maintenance scope, engine configuration, or registration structure. The result may also reflect that the two vessel types are not directly comparable in every respect.

Another important finding is that recurring LTSA-related entries must be treated carefully. If all recurring LTSA rows are treated as duplicate purchase-order versions, the Passenger Vessel's procurement exposure is understated. If every repeated purchase-order row is counted without classification, there is a risk of double-counting ordinary version updates. The corrected model therefore improves the analysis by distinguishing between ordinary purchase-order version updates and recurring agreement-related entries.

The availability finding was also less useful than expected. One possible expectation could be that the LTSA-based case would show stronger availability because of structured support. However, the BASSnet downtime field does not provide usable positive downtime values for either vessel. Therefore, the analysis cannot conclude that either arrangement had better real availability. This is an important finding because it shows that availability cannot be evaluated reliably unless downtime and operational consequence data are recorded consistently.

9.5 Practical implications for Wilhelmsen Ship Management

The findings have practical relevance for Wilhelmsen Ship Management because they show that maintenance arrangement evaluation should be based on several indicators at the same time. A contract structure alone does not determine whether a strategy is better. The evaluation should include procurement exposure, spare-parts support, maintenance activity, availability indicators, financial risk exposure, and data quality.

For Wilhelmsen Ship Management, one implication is that LTSA-based arrangements should be assessed as both service agreements and procurement-exposure structures. The Passenger Vessel case shows that significant procurement-related exposure can remain visible even under an LTSA-based arrangement. This means that future LTSA evaluations should clarify which maintenance activities, spare parts, refurbishment work, and technical services are included in the agreement, and which remain outside the agreement.

A second implication is that non-LTSA arrangements may require stronger internal spare-parts and procurement control. The Heavy Lift Vessel has much higher visible spare-parts stock exposure in the available BASSnet data. This suggests that non-LTSA arrangements may require more attention to inventory planning, supplier management, lead times, repairable components, and refurbishment logistics.

A third implication concerns procurement-data classification. The corrected analysis shows that recurring agreement-related entries must be separated from ordinary purchase-order version updates. Without this distinction, procurement exposure can be misinterpreted. For future analysis, Wilhelmsen Ship Management should consider using clearer classification fields for recurring agreement costs, ordinary purchase orders, cancelled orders, version updates, long-lead items, and service-related procurement.

A fourth implication concerns data quality. The analysis was limited by unequal data availability between the two vessel cases. The Heavy Lift Vessel had defect records, while the Passenger Vessel did not have a comparable defect dataset. The downtime field was also not useful for a full availability comparison. If Wilhelmsen Ship Management wants stronger future analysis, it would be valuable to standardize how maintenance events, defects, downtime, purchase orders, spare-parts usage, and agreement-related costs are recorded.

A final implication is that BASSnet data can support maintenance-strategy evaluation, but only if the data is interpreted correctly. Job-history records should not automatically be treated as failures, purchase-order values should not automatically be treated as complete cost, and spare-parts records should not automatically be treated as consumption. Used carefully, however, these data categories can provide a useful decision-support framework.

9.6 Critical assessment of the method

The method used in this thesis has several strengths. First, it is based on real company data rather than only theoretical assumptions. Second, it combines several types of operational evidence, including job-history records, purchase-order records, spare-parts records, and defect records where available. Third, it uses Monte Carlo simulation and sensitivity analysis to account for uncertainty instead of relying only on one deterministic value.

The method is also transparent. The thesis explains how purchase-order records are filtered, why cancelled orders are excluded, why recurring LTSA-related entries are retained separately, why spare-parts records are treated as stock data rather than consumption data, and why availability is treated as a proxy rather than as a complete technical measure. These choices make the analysis easier to reproduce and easier to critique.

At the same time, the method has weaknesses. The two cases are not fully symmetrical. The Passenger Vessel and Heavy Lift Vessel differ in vessel type, operational profile, maintenance structure, engine configuration, and data availability. The Heavy Lift Vessel has defect data, while the Passenger Vessel does not. This means that defect frequency cannot be directly compared.

Another weakness is that the model does not include all possible cost or performance elements. Purchase-order values are useful indicators of procurement-related exposure, but they do not capture every maintenance-related cost. The model also does not reproduce the full technical behavior of the engine systems. It is therefore best understood as a simplified decision-support framework rather than a complete financial or engineering model.

The corrected procurement method also depends on classification judgement. The distinction between ordinary purchase-order version updates and recurring agreement-related entries is necessary, but it requires interpretation of descriptions and registration patterns. This is a strength because it avoids understating LTSA exposure, but it is also a limitation because it depends on the quality and consistency of the available records.

The availability analysis is the weakest part of the method because the available downtime field does not provide useful positive downtime observations. This does not make the thesis invalid, but it means that the operational availability part of the research question can only be answered cautiously. The thesis can discuss availability as a limitation and decision-relevant dimension, but it cannot prove which arrangement produced better real availability.

9.7 Reliability, validity, and generalizability

The reliability of the study depends on the consistency of the extracted data and the transparency of the modelling process. Since the same final BASSnet CSV files and the same calculation rules can be used to reproduce the analysis, the internal reliability of the calculations is relatively strong. The use of explicit assumptions, fixed filtering logic, corrected procurement classification, and Monte Carlo simulation also supports reliability.

Validity is more complex. The study has practical validity because it addresses a real maintenance arrangement problem using real company data. It compares two actual vessel cases and two actual maintenance arrangements. However, the model is still a simplified representation of reality. It does not include every technical, contractual, operational, or financial factor that could influence the full value of an LTSA-based or non-LTSA arrangement.

The validity of the procurement-related cost exposure dimension is strongest when interpreted as visible procurement exposure, not as complete cost. The corrected classification improves validity because it avoids understating recurring LTSA-related exposure. The validity of the availability dimension is weaker because the downtime field does not provide usable positive downtime values. The validity of the financial risk dimension is stronger because the purchase-order values can be used to analyze variation, concentration, uncertainty, and exposure.

Generalizability is limited. The thesis is based on two selected vessels, one company context, and one selected two-year period. The findings cannot be generalized statistically to all vessels, all engine types, or all LTSA-based and non-LTSA arrangements. However, the thesis has analytical generalizability. The framework, reasoning, and modelling approach may be useful in other maintenance arrangement evaluations where companies need to compare contract-based maintenance with direct maintenance management under data limitations.

9.8 Limitations and weaknesses of the study

The main limitation of the study is that the comparison is based on two selected vessel cases rather than a large sample of vessels or maintenance arrangements. The results are therefore case-specific and should not be interpreted as statistically representative of all maritime maintenance arrangements.

The selected study period also limits the strength of the conclusions. The analysis covers 2024–2025, which provides a recent and manageable dataset, but it may not capture longer maintenance cycles, major overhaul timing, unusual cost years, or changes in contract performance over time. A longer observation period could strengthen the comparison and might produce different results.

Data availability is another important limitation. The Heavy Lift Vessel has defect records, while the Passenger Vessel does not have a comparable defect dataset in the final CSV files. This means that defect frequency cannot be compared directly between the two cases. The analysis therefore relies more heavily on job-history records, purchase-order records, and spare-parts records.

The availability result is especially uncertain because the available downtime field shows zero positive recorded downtime hours for both vessels. This should not be interpreted as proof of perfect operational availability. It only means that the selected BASSnet downtime field does not provide sufficient evidence for a complete availability comparison.

Finally, the model depends on assumptions about procurement exposure, spare-parts stock value, and uncertainty. These assumptions are necessary because not all relevant variables are directly observable in the available data. Sensitivity analysis helps show whether the main findings are robust, but it does not remove the underlying uncertainty. The results should therefore be understood as structured decision-support findings rather than exact engineering or accounting measurements.

9.9 Summary

This chapter has discussed the findings in relation to the research question, theory, practical implications, and methodological limitations. The results show that, within the available 2024–2025 BASSnet data and the corrected modelling framework, the Passenger Vessel has higher visible procurement-related cost exposure and higher recorded maintenance activity, while the Heavy Lift Vessel has higher visible spare-parts stock exposure.

The Monte Carlo simulation supports the finding that the Passenger Vessel has higher procurement-related exposure under the model assumptions. However, the results do not prove that LTSA-based arrangements are generally more expensive or that non-LTSA arrangements are generally better. The two cases differ in vessel type, engine configuration, operational profile, agreement structure, and data availability.

The availability result is inconclusive because the available downtime field does not contain usable positive downtime values for either vessel case. The defect comparison is also limited because defect records are only available for the Heavy Lift Vessel. The practical value of the thesis is therefore not only in ranking the two cases, but in showing how BASSnet data can be structured into a decision-support framework for comparing maintenance arrangements under real data limitations.

10.0 Conclusion

This thesis has examined how an LTSA-based maintenance arrangement compares with a non-LTSA maintenance arrangement for MAN marine engines using available BASSnet records. The study compared two anonymized vessel cases within the Wilhelmsen Ship Management portfolio: the Passenger Vessel, representing the LTSA-based case, and the Heavy Lift Vessel, representing the non-LTSA case. The analysis was based only on engine-related BASSnet CSV data from the 2024–2025 period, including job-history records, purchase-order records, spare-parts records, and defect records where available.

The main conclusion is that the two maintenance arrangements create different types of operational and financial exposure. Within the available BASSnet data, the Passenger Vessel has higher visible procurement-related cost exposure and higher recorded maintenance activity, while the Heavy Lift Vessel has higher visible spare-parts stock exposure. This means that the comparison cannot be reduced to a simple statement that one strategy is better than the other. Instead, the results show that LTSA-based and non-LTSA arrangements distribute maintenance responsibility, procurement exposure, spare-parts support, and uncertainty in different ways.

For procurement-related cost exposure, the Passenger Vessel shows a substantially higher value than the Heavy Lift Vessel after correcting the purchase-order classification. The corrected purchase-order exposure indicator is USD 2,174,227.72 for the Passenger Vessel and USD 454,868.63 for the Heavy Lift Vessel. This gives a corrected procurement exposure ratio of 4.78, meaning that the Passenger Vessel has approximately 4.78 times higher visible procurement-related cost exposure than the Heavy Lift Vessel in the available BASSnet purchase-order records. This result is based on separating ordinary purchase-order version updates from recurring LTSA-related agreement entries, so that recurring monthly, running-hour, and long-lead agreement entries are not incorrectly removed as duplicate purchase-order versions.

The Monte Carlo simulation supports this result. The mean simulated procurement-related exposure is USD 2,169,870.79 for the Passenger Vessel and USD 454,327.73 for the Heavy Lift Vessel. The paired simulation shows that the Passenger Vessel has higher simulated procurement-related exposure in 100.00% of the simulation runs. This indicates that the procurement-exposure result is robust within the assumptions of the model. However, it should not be interpreted as proof that LTSA-based arrangements are generally more expensive than non-LTSA arrangements. The result is specific to the two vessel cases, the selected time period, the available BASSnet records, and the modelling assumptions used in the thesis.

The spare-parts results show a different pattern. The Heavy Lift Vessel has a spare-parts stock-value indicator of USD 244,603.78, while the Passenger Vessel has a spare-parts stock-value indicator of USD 7,804.87. This gives a spare-parts stock-value ratio of 31.34. This does not mean that the Heavy Lift Vessel consumed more spare parts during the study period. It means that the visible spare-parts stock and support structure is larger in the available BASSnet data. This may reflect that the non-LTSA case carries more visible internal responsibility for spare-parts planning, procurement, stockholding, and availability support.

For maintenance activity, the Passenger Vessel has 2,691 job-history records, compared with 1,062 for the Heavy Lift Vessel. The Passenger Vessel also has a higher overdue job share, at 24.45%, compared with 9.89% for the Heavy Lift Vessel. These results indicate more recorded maintenance activity and follow-up pressure in the available BASSnet data for the Passenger Vessel. However, job-history records should not be interpreted directly as failure records. They may include planned maintenance, inspections, service tasks, administrative follow-up, corrective work, and other registered activities. Differences in record counts may also reflect vessel type, engine configuration, operational profile, agreement structure, registration practice, or how jobs are divided in BASSnet.

For operational availability, the conclusion is less definitive. The selected BASSnet downtime field shows zero positive recorded downtime hours for both vessel cases. This gives an availability proxy of 100% for both vessels, but this should not be interpreted as proof that both vessels had perfect operational availability. It only shows that the available downtime field does not provide usable positive downtime values for the selected engine-related records. Therefore, the thesis cannot conclude which maintenance arrangement provides better real operational availability.

The defect data also limits the comparison. The Heavy Lift Vessel has 99 defect records, while the Passenger Vessel does not have a comparable defect dataset in the final CSV files. The Heavy Lift Vessel defect data can therefore support interpretation of that case, but it cannot be used for a direct defect-frequency comparison between the two arrangements. The absence of Passenger Vessel defect records does not mean that no defects occurred. It only means that comparable defect data was not available in the final data foundation.

For Wilhelmsen Ship Management, the practical implication is that maintenance strategies should be evaluated using several indicators at the same time. A contract structure alone does not determine whether a strategy is better. LTSA-based arrangements may provide structure, predictability, technical support, and clearer responsibility, but they do not automatically remove all procurement exposure or maintenance-related cost. Non-LTSA arrangements may provide more direct control, but they may also require stronger internal spare-parts planning, supplier follow-up, procurement management, and refurbishment coordination.

The thesis also demonstrates the value of structured internal data analysis. BASSnet records can support maintenance-strategy evaluation when they are interpreted correctly. Job-history records should not automatically be treated as failures, purchase-order values should not be treated as complete vessel cost, and spare-parts records should not be treated as consumption history. Used carefully, these records can form the basis for a transparent decision-support framework.

At the same time, the conclusion must be interpreted in light of the study’s limitations. The analysis is based on two selected vessel cases, one company context, and a two-year data period. The two vessels differ in vessel type, operational profile, engine configuration, agreement structure, and data availability. The model is also simplified and does not include every technical, contractual, operational, or financial factor that could influence the full value of an LTSA-based or non-LTSA arrangement. The findings are therefore case-specific and should not be interpreted as universal evidence that one maintenance arrangement is generally superior.

Further research could strengthen the analysis by including more vessels, a longer time horizon, more complete downtime records, comparable defect data for both vessel cases, and clearer links between purchase orders, spare-parts consumption, maintenance jobs, and agreement coverage. It would also be valuable to compare several LTSA-based and non-LTSA arrangements to examine how pricing, service scope, spare-parts responsibility, refurbishment obligations, and performance guarantees influence procurement exposure, spare-parts support, and operational performance.

In conclusion, this thesis has shown that a structured comparison of LTSA-based and non-LTSA maintenance arrangements is both possible and valuable under real data limitations. Within the available BASSnet data, the LTSA-based case has higher visible procurement-related cost exposure and higher recorded maintenance activity, while the non-LTSA case has higher visible spare-parts stock exposure. The availability comparison remains inconclusive. The broader contribution of the thesis is therefore not only the comparison of the two cases, but the development of a transparent analytical framework for evaluating maintenance strategy using operational and procurement data.

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12.0 Appendices

The appendices provide supporting documentation for the data preparation, modelling, Monte Carlo simulation, sensitivity analysis, and additional figures used in the thesis. The raw company exports are not reproduced in full because they may contain confidential internal information. Instead, the appendices document the cleaned and structured BASSnet CSV data, the interpretation rules, the model outputs, and the Python code used to reproduce the calculations.

12.1 Appendix A: Overview of BASSnet CSV Files

Appendix A provides an overview of the final BASSnet CSV files used as the empirical data foundation for the thesis. The final analysis is based only on engine-related BASSnet records for the 2024–2025 period. The two vessel cases are anonymised as Passenger Vessel and Heavy Lift Vessel.

The CSV files contain job-history records, purchase-order records, spare-parts records, and defect records where available. The files were cleaned and structured before analysis so that they could support the record-count tables, procurement cost exposure indicators, spare-parts indicators, Monte Carlo simulation, and sensitivity analysis presented in Chapters 7 and 8.

Table A.1: Overview of final BASSnet CSV files used in the thesis

Data file/category	Vessel case	Strategy	Main content	Role in thesis
Job-history CSV	Passenger Vessel	LTSA	Engine-related job-history records for 2024–2025	Used to analyse recorded maintenance activity and availability proxy
Purchase-order CSV	Passenger Vessel	LTSA	Engine-related purchase-order records	Used to analyse procurement-related cost exposure in USD
Spare-parts CSV	Passenger Vessel	LTSA	Spare-parts master, stock, and average-price records	Used to analyse spare-parts support structure and stock-value exposure
Job-history CSV	Heavy Lift Vessel	Non-LTSA	Engine-related job-history records for 2024–2025	Used to analyse recorded maintenance activity and availability proxy
Purchase-order CSV	Heavy Lift Vessel	Non-LTSA	Engine-related purchase-order records	Used to analyse procurement-related cost exposure in USD
Spare-parts CSV	Heavy Lift Vessel	Non-LTSA	Spare-parts master, stock, and average-price records	Used to analyse spare-parts support structure and stock-value exposure
Defect CSV	Heavy Lift Vessel	Non-LTSA	Defect records for 2024–2025	Used as supporting defect evidence for the Heavy Lift Vessel case only

Source: Prepared by Evan Elias Håpnes based on final engine-related BASSnet CSV files from Wilhelmsen Ship Management.

The final BASSnet CSV files were used as the only empirical data source in the revised thesis. Job-history records were interpreted as maintenance activity records, purchase-order records were interpreted as procurement-related cost exposure indicators, spare-parts records were interpreted as stock and support-structure data, and defect records were used only where available.

12.2 Appendix B: Data cleaning and classification rules

Appendix B documents the main data cleaning and classification rules used when preparing the BASSnet CSV files for analysis. The purpose of these rules was to make the data suitable for a structured comparison of the LTSA case and the non-LTSA case while avoiding misleading interpretations.

The original data was extracted from BASSnet and then reduced to the records relevant for the thesis question. The cleaning process focused on the 2024–2025 period, the two anonymized vessel cases, engine-related records, and the model categories used in Chapters 6, 7, and 8.

Table B.1: Data cleaning and classification rules

Area	Cleaning or classification rule	Reason for rule
Thesis period	Records were limited to the 2024–2025 analysis period where possible	This created a common comparison period for both vessel cases
Vessel anonymization	Vessel names were replaced with Passenger Vessel and Heavy Lift Vessel	This follows confidentiality requirements and avoids client-specific naming
Strategy classification	Passenger Vessel was classified as the LTSA case, while Heavy Lift Vessel was classified as the non-LTSA case	This matched the research design and allowed the two maintenance structures to be compared
Source selection	Only final BASSnet CSV files were used in the revised analysis	This ensures that the thesis is based on the corrected operational and procurement dataset
Job-history records	Job-history records were treated as evidence of recorded maintenance activity, not as direct failure counts	Job records may include planned work, inspections, service tasks, corrective work, or administrative follow-up
Corrective jobs	Corrective job records were identified using the Job Class field	This provides a consistent classification rule across the job-history data
Overdue jobs	Overdue records were counted from the available overdue/status fields in the job-history data	This supports analysis of maintenance planning and follow-up pressure
Purchase-order rows	Purchase-order rows were not treated as unique purchase orders without filtering	The same purchase order can appear in several rows due to versions, updates, or line items

Purchase-order versions	Only the latest version of each purchase order was retained for the procurement-exposure model	This reduces the risk of double-counting procurement values
Cancelled purchase orders	Cancelled purchase orders were excluded from the Monte Carlo cost simulation	Cancelled orders do not represent completed procurement exposure in the same way as active or received orders
Monetary values	All monetary values were treated as USD	This matches the currency used in the final BASSnet CSV files
Spare-parts records	Spare-parts records were interpreted as stock and support-structure data, not direct consumption history	Spare-parts master records do not prove that listed parts were consumed during the period
Spare-parts stock value	Stock-value indicators were calculated as stock quantity multiplied by average price	This gives a visible stock-exposure indicator in USD
Defect records	Defect records were used only for the Heavy Lift Vessel case	Comparable defect data was not available for the Passenger Vessel
Downtime records	Downtime was treated as an availability proxy only	The available downtime field does not provide a complete technical availability measure
Missing values	Missing values were not treated as zero unless zero was clearly supported by the data field	This avoids creating false results from missing data
Monte Carlo simulation	The simulation used filtered purchase-order values from BASSnet	This links the stochastic model directly to the final data foundation
Sensitivity testing	Sensitivity analysis tested purchase-order exposure, spare-parts weighting, and downtime scenarios	This tested whether the interpretation depends strongly on uncertain assumptions
Purchase-order version identifiers	Ordinary purchase-order version updates were grouped by purchase-order identity, including version indicators such as .ver. where present. Only the latest non-cancelled ordinary version was retained for ordinary purchase-order exposure.	This reduces double-counting caused by repeated purchase-order versions, updates, or registrations.
Recurring LTSA-related entries	Recurring monthly, running-hour, long-lead, and agreement-related LTSA entries were retained separately when the description and registration pattern indicated distinct recurring cost exposure rather than duplicate versions of the same order.	This avoids understating the visible LTSA-related procurement exposure for the Passenger Vessel.
Monetary value cleaning	Monetary fields were cleaned before calculation by treating comma-formatted values as thousands-separated USD values and preserving decimal notation.	This prevents numeric conversion errors when values such as 1,359,948.51 are imported from CSV files.
Corrected procurement exposure	Corrected procurement exposure was calculated as ordinary latest-version non-cancelled purchase-order exposure plus retained recurring LTSA-related agreement exposure.	This gives a more realistic procurement-related cost exposure indicator than a pure latest-version-only rule.

The most important cleaning rule was that different BASSnet record types were not interpreted as if they represented the same kind of evidence. Job-history records describe recorded maintenance activity, purchase-order records describe procurement exposure, spare-parts records describe stock and support structure, and defect records describe technical issues only where defect data is available.

12.3 Appendix C: Extracted BASSnet record-count and indicator tables

Appendix C presents the extracted record-count and indicator tables used to support the empirical data overview in Chapter 8. These tables summarize the final BASSnet data foundation after cleaning and classification.

Table C.1: BASSnet record counts used in the thesis

Vessel case	Strategy	Dataset	Total records
Passenger Vessel	LTSA	Job history	2,691
Passenger Vessel	LTSA	Purchase orders	134
Passenger Vessel	LTSA	Spare parts	104
Passenger Vessel	LTSA	Defects	Not available
Heavy Lift Vessel	Non-LTSA	Job history	1,062
Heavy Lift Vessel	Non-LTSA	Purchase orders	83
Heavy Lift Vessel	Non-LTSA	Spare parts	121
Heavy Lift Vessel	Non-LTSA	Defects	99

Source: Prepared by Evan Elias Håpnes based on final engine-related BASSnet CSV files.

The record counts show that the two cases are not fully symmetrical. Both vessel cases have job-history, purchase-order, and spare-parts records, but only the Heavy Lift Vessel has defect records in the final data foundation. This means that defect-level analysis can support the Heavy Lift Vessel case, but direct defect-frequency comparison between the two cases is not possible.

Table C.2: Maintenance activity indicators

Indicator	Passenger Vessel, LTSA	Heavy Lift Vessel, non-LTSA
Job-history records	2,691	1,062
Corrective job records	59	1
Corrective job share	2.19%	0.09%
Overdue job records	658	105
Overdue job share	24.45%	9.89%
Positive recorded downtime hours	0	0

Source: Prepared by Evan Elias Håpnes based on final BASSnet job-history CSV files.

Job-history records are used as evidence of recorded maintenance activity, not as evidence of failure frequency. A higher number of job-history records may reflect more recorded work, but it may also reflect differences in planned maintenance structure, vessel type, operational profile, or registration practice.

Table C.3: Purchase-order indicators

Indicator	Passenger Vessel, LTSA	Heavy Lift Vessel, non-LTSA
Purchase-order rows	134	83
Ordinary latest-version, non-cancelled purchase orders	50	69
Recurring agreement-related entries retained separately	66	0
LTSA running-hour/monthly exposure, USD	1,359,948.51	Not applicable
Long-lead/Buyersclub recurring exposure, USD	58,332.24	Not applicable
Ordinary purchase-order exposure, USD	755,946.97	454,868.63
Corrected purchase-order exposure indicator, USD	2,174,227.72	454,868.63
Mean corrected purchase-order event value, USD	18,743.34	6,592.30
Standard deviation of corrected purchase-order event value, USD	22,204.15	9,045.92
Coefficient of variation	1.18	1.37
Top-five purchase-order concentration	20.74%	35.91%

Source: Prepared by Evan Elias Håpnes based on final BASSnet purchase-order CSV files.

The purchase-order indicators are based on corrected procurement classification. Ordinary purchase-order version updates are handled using the latest non-cancelled version, while recurring LTSA-related monthly, running-hour, and long-lead agreement entries are retained separately when they represent distinct recurring exposure rather than duplicate purchase-order versions. The values are used as procurement-related cost exposure indicators, not as complete vessel lifecycle cost values.

Table C.4: Spare-parts indicators

Indicator	Passenger Vessel, LTSA	Heavy Lift Vessel, non-LTSA
Spare-parts records	104	121
Spare-parts records with stock above zero	16	112
Total stock quantity	30	777
Spare-parts stock-value indicator, USD	7,804.87	244,603.78

Source: Prepared by Evan Elias Håpnes based on final BASSnet spare-parts CSV files.

Spare-parts records are interpreted as stock and support-structure data. They are not treated as evidence that the listed parts were consumed during 2024–2025.

Table C.5: Heavy Lift Vessel defect record overview

Defect indicator	Heavy Lift Vessel
Total defect records	99

Closed defect records	79
Completed defect records	10
New defect records	10
Overdue defect records	5
Critical defect records	24
Stand-by critical defect records	13
Non-critical defect records	54
Records without priority classification	8

Source: Prepared by Evan Elias Håpnes based on final BASSnet defect CSV file.

The defect table is included as supporting information for the Heavy Lift Vessel case. It cannot be used for direct comparison with the Passenger Vessel because comparable defect records were not available for the Passenger Vessel.

12.4 Appendix D: Monte Carlo and Sensitivity output

Appendix D presents the main numerical outputs from the Monte Carlo simulation and sensitivity analysis used in Chapter 8. The purpose of this appendix is to document the model outputs without reproducing all 10,000 simulation rows.

The simulation uses latest-version, non-cancelled BASSnet purchase-order values. It does not estimate complete vessel lifecycle cost. Instead, it simulates possible variation in procurement-related cost exposure visible in the available engine-related purchase-order records. All monetary values are presented in USD.

Table D.1: Monte Carlo procurement-exposure summary

Monte Carlo output	Passenger Vessel, LTSA	Heavy Lift Vessel, non-LTSA
Mean simulated exposure, USD	2,169,870.79	454,327.73
Median simulated exposure, USD	2,151,475.36	449,822.85
Standard deviation, USD	207,780.37	92,630.63
5th percentile, USD	1,857,507.98	310,961.59
95th percentile, USD	2,537,313.49	616,492.15

Source: Prepared by Evan Elias Håpnes based on the BASSnet-based Monte Carlo model.

The Monte Carlo output shows that the Passenger Vessel has higher simulated procurement-related cost exposure than the Heavy Lift Vessel under the corrected model assumptions. The simulation separates recurring LTSA-related agreement exposure from ordinary purchase-order exposure.

Table D.2 Paired Monte Carlo difference summary

Metric	Result
Mean Passenger Vessel minus Heavy Lift Vessel exposure, USD	1,715,543.07
5th percentile difference, USD	1,371,638.80
95th percentile difference, USD	2,109,030.81
Probability that Passenger Vessel exposure is higher	100.00%
Probability that Heavy Lift Vessel exposure is higher	0.00%

Source: Prepared by Evan Elias Håpnes based on the BASSnet-based Monte Carlo model.

The paired simulation shows that the Passenger Vessel has higher simulated procurement-related cost exposure in all simulation runs. This supports the deterministic procurement-exposure result, but it should still be interpreted as procurement-related exposure only, not as a complete vessel-cost conclusion.

Table D.3: Sensitivity analysis summary

Sensitivity test	Result	Interpretation
Heavy Lift Vessel purchase-order break-even factor	4.78	Heavy Lift Vessel purchase-order exposure would need to increase by approximately 378% to equal the Passenger Vessel's corrected purchase-order exposure.
Passenger Vessel purchase-order reduction factor	0.21	Passenger Vessel purchase-order exposure would need to fall to about 20.9% of its corrected observed value to equal the Heavy Lift Vessel.
Spare-parts stock-value break-even factor	7.26	Heavy Lift Vessel spare-parts stock value must be weighted very heavily before it overturns the corrected purchase-order exposure difference.
Recorded downtime sensitivity	Inconclusive	The available downtime field contains zero positive recorded downtime hours for both cases.

Source: Prepared by Evan Elias Håpnes based on the BASSnet-based Monte Carlo model.

The sensitivity analysis shows that the corrected procurement-exposure result is robust under the model assumptions. The availability result remains weaker because the available downtime field does not provide usable positive downtime observations.

12.5 Appendix E: Python simulation code

The Python script in this appendix was developed with support from artificial intelligence (AI) tools. The assistance included code structuring, syntax support, debugging, Monte Carlo simulation logic, sensitivity analysis, and reproducibility checks. The script was tested, adjusted, and verified by Evan Elias Håpnes before being used in the thesis analysis. The use of AI tools did not replace the author's responsibility for the calculations, interpretations, or final results.

Appendix E presents the Python script used to reproduce the deterministic indicators, Monte Carlo simulation, sensitivity checks, and reproducibility output files used in the thesis. The script reads the anonymized thesis calculation workbook, which was prepared from the cleaned BASSnet CSV files described in Appendix B and Appendix C. It recalculates the deterministic indicators used in Chapters 7 and 8, reruns the Monte Carlo procurement-exposure model, compares the recalculated Monte Carlo output with the earlier reported values, and exports auditable output tables.

The raw company exports and original file names are not reproduced in full because they may contain confidential internal information, including vessel-specific names, system labels, operational identifiers, and supplier-related information. For this reason, the workbook, vessel labels, and displayed output labels have been anonymized to match the case names used in the thesis: Passenger Vessel and Heavy Lift Vessel. The anonymization changes only the displayed file names and labels. It does not change the calculation logic, data-cleaning rules, or model structure.

The executable Python file should be kept separately together with the anonymized thesis calculation workbook. The code is included here to document the reproducibility process, but the separate .py file should be used for execution because formatting and indentation may not be preserved perfectly in the PDF version of the appendix.

Code E.1: Python reproducibility script:

```
"""

Fixed reproducibility script for the anonymized LOG650 thesis calculation dataset.


Purpose

-------

This script is intended to replace/clean up Appendix E for reproducibility.

It reads the anonymized mother Excel workbook, recalculates the deterministic

BASSnet indicators, reruns the Monte Carlo model with deterministic input ordering,

and writes auditable output files.


Main fixes compared with the earlier Appendix E logic
```

----

1. Uses RuntimeError instead of an invalid UnicodeDecodeError constructor.

2. Uses deterministic ordering of Monte Carlo input events.

3. Uses independent random-number streams for Passenger Vessel and Heavy Lift Vessel.

4. Saves input-file hash, deterministic checks, Monte Carlo samples, Monte Carlo summary,

   and a reproducibility audit note.

5. Does not silently claim exact reproduction of old Monte Carlo values if the anonymized

   workbook only contains rounded values or lacks the original Monte Carlo input order.

Requirements

----

pip install pandas numpy openpyxl

How to run

----

1. Place this script in the same folder as the anonymized mother workbook.

2. Make sure the workbook is named either:

      Anonymized_Thesis_Calculation_Dataset.xlsx

   or:

      Anonymized_Thesis_Calculation_Dataset(1).xlsx

   You can also edit MOTHER_WORKBOOK below.

3. Run:

      reproducibility_monte_carlo.py

Output folder

```
-------------
reproducibility_output/
"""

from __future__ import annotations

from dataclasses import dataclass
from pathlib import Path
import hashlib
import platform
import re
import warnings

import numpy as np
import pandas as pd

# ============================================================
# Reproducibility settings
# ============================================================

RANDOM_SEED = 42
SIM_RUNS = 10_000
OUTPUT_DIR = Path("reproducibility_output")

# Set to True only if you have added a true original Monte Carlo order column
```

```
# and full-precision event values, and you expect the old thesis MC table to match exactly.
STRICT_REPORTED_MC_VALIDATION = False

# Leave as None to auto-detect the workbook in the current folder.
MOTHER_WORKBOOK: Path | None = None

WORKBOOK_CANDIDATES = [
    Path("Anonymized_Thesis_Calculation_Dataset.xlsx"),
    Path("Anonymized_Thesis_Calculation_Dataset(1).xlsx"),
]

PASSENGER = "Passenger Vessel, LTSA"
HEAVY = "Heavy Lift Vessel, non-LTSA"

# Deterministic thesis values to validate.
EXPECTED_DETERMINISTIC = {
    "passenger_job_records": 2691,
    "heavy_job_records": 1062,
    "passenger_corrective_jobs": 59,
    "heavy_corrective_jobs": 1,
    "passenger_overdue_jobs": 658,
    "heavy_overdue_jobs": 105,
    "passenger_po_rows": 134,
    "heavy_po_rows": 83,
    "passenger_ordinary_po_count": 50,
```

```
    "heavy_ordinary_po_count": 69,
    "passenger_recurring_entries": 66,
    "heavy_recurring_entries": 0,
    "passenger_recurring_exposure": 1_359_948.51,
    "passenger_long_lead_exposure": 58_332.24,
    "passenger_ordinary_exposure": 755_946.97,
    "heavy_ordinary_exposure": 454_868.63,
    "passenger_corrected_exposure": 2_174_227.72,
    "heavy_corrected_exposure": 454_868.63,
    "passenger_spare_records": 104,
    "heavy_spare_records": 121,
    "passenger_stock_above_zero": 16,
    "heavy_stock_above_zero": 112,
    "passenger_total_stock_qty": 30,
    "heavy_total_stock_qty": 777,
    "passenger_stock_value": 7_804.8665,
    "heavy_stock_value": 244_603.7789,
    "heavy_defect_records": 99,
}


# Old reported Monte Carlo table values from the thesis/mother workbook.
# These are kept for comparison only. The corrected script recalculates MC output
# from the event data. Small differences can occur if anonymized event values are rounded
# or if the original MC input order is not included in the anonymized workbook.
REPORTED_MC = {
```

```python
    "passenger_mean": 2_169_870.79,
    "passenger_median": 2_151_475.36,
    "passenger_std": 207_780.37,
    "passenger_p5": 1_857_507.98,
    "passenger_p95": 2_537_313.49,
    "heavy_mean": 454_327.73,
    "heavy_median": 449_822.85,
    "heavy_std": 92_630.63,
    "heavy_p5": 310_961.59,
    "heavy_p95": 616_492.15,
    "diff_mean": 1_715_543.07,
    "diff_p5": 1_371_638.80,
    "diff_p95": 2_109_030.81,
}

# ============================================================
# Helper functions
# ============================================================


def locate_workbook() -> Path:
    """Find the anonymized mother workbook."""
    if MOTHER_WORKBOOK is not None:
        if not MOTHER_WORKBOOK.exists():
            raise FileNotFoundError(f"Workbook not found: {MOTHER_WORKBOOK}")
        return MOTHER_WORKBOOK
```

```
    for candidate in WORKBOOK_CANDIDATES:
        if candidate.exists():
            return candidate

    raise FileNotFoundError(
        "Could not find the anonymized mother workbook. Place it next to this script "
        "or edit MOTHER_WORKBOOK in the script."
    )


def file_hash(path: Path) -> str:
    """Create a SHA-256 hash so the exact input file can be verified."""
    h = hashlib.sha256()
    with path.open("rb") as f:
        for block in iter(lambda: f.read(8192), b""):
            h.update(block)
    return h.hexdigest()


def read_csv_reproducible(path: Path) -> pd.DataFrame:
    """
    Read CSV files using common encodings for exported operational-system data.

    This fixes the earlier Appendix E issue where UnicodeDecodeError was constructed
    incorrectly. RuntimeError is the correct simple exception here.
    """
```

```
    encodings = ["utf-8-sig", "cp1252", "latin1"]
    last_error: Exception | None = None

    for enc in encodings:
        try:
            return pd.read_csv(path, encoding=enc)
        except UnicodeDecodeError as exc:
            last_error = exc
            continue

    raise RuntimeError(
        f"Could not read {path} with utf-8-sig, cp1252, or latin1 encoding."
    ) from last_error


def normalize_text(value: object) -> str:
    """Normalize text for robust comparisons."""
    return str(value).strip().lower()


def normalize_col_name(name: object) -> str:
    """Normalize column names for flexible matching."""
    return (
        str(name)
        .lower()
        .replace(".", "")
        .replace("_", " ")
```

```
        .replace("-", " ")
        .strip()
    )


def true_like(series: pd.Series) -> pd.Series:
    """Interpret common yes/no and true/false values."""
    return series.astype(str).str.lower().str.strip().isin(["yes", "y", "true", "1", "x"])


def money_to_float(series: pd.Series) -> pd.Series:
    """Convert monetary values to float while handling commas, blanks, and
parentheses."""
    cleaned = (
        series.astype(str)
        .str.replace(",", "", regex=False)
        .str.replace("(", "-", regex=False)
        .str.replace(")", "", regex=False)
        .str.strip()
        .replace(
            {
                "": "0",
                "nan": "0",
                "None": "0",
                "NaN": "0",
                "Not applicable": "0",
                "not applicable": "0",
            }
```

```
        )
    )
    return pd.to_numeric(cleaned, errors="coerce").fillna(0.0)


def assert_close(name: str, actual: float, expected: float, tolerance: float = 0.01) -> None:
    """Fail loudly if a calculated value does not match the thesis/mother-doc value."""
    if abs(float(actual) - float(expected)) > tolerance:
        raise AssertionError(
            f"{name} mismatch: actual={actual:.6f}, expected={expected:.6f}, "
            f"difference={actual - expected:.6f}"
        )


def event_id_number(value: object) -> int:
    """Extract the trailing number from IDs such as PV_PO_001 or HL_PO_069."""
    match = re.search(r"(\d+)\s*$", str(value))
    if match:
        return int(match.group(1))
    return 10**9


def read_sheet_table(workbook_path: Path, sheet_name: str) -> pd.DataFrame:
    """Read a normal worksheet with headers in row 1."""
    return pd.read_excel(workbook_path, sheet_name=sheet_name)


def read_summary_sheet(workbook_path: Path, sheet_name: str, header_label: str) ->
pd.DataFrame:
    """
```

```
    Read a summary worksheet where the actual header row appears below title/blank rows.

    Example header labels: 'Metric', 'Indicator', 'Defect indicator'.

    """

    raw = pd.read_excel(workbook_path, sheet_name=sheet_name, header=None)

    header_row_candidates = raw.index[raw.iloc[:,
0].astype(str).str.strip().eq(header_label)].tolist()

    if not header_row_candidates:

        raise ValueError(f"Could not find header label '{header_label}' in sheet
'{sheet_name}'.")



    header_row = header_row_candidates[0]

    headers = raw.iloc[header_row].tolist()

    data = raw.iloc[header_row + 1 :].copy()

    data.columns = headers

    data = data.dropna(how="all")

    return data.reset_index(drop=True)


def metric_value(summary: pd.DataFrame, label_col: str, metric: str, value_col: str) ->
object:

    """Get one value from a summary table by row label and column label."""

    rows = summary[summary[label_col].astype(str).str.strip().eq(metric)]

    if rows.empty:

        raise KeyError(f"Metric not found: {metric}")

    return rows.iloc[0][value_col]


# ===========================================================

# Deterministic calculations from the mother workbook
```

```
# ============================================================

@dataclass(frozen=True)

class POResults:

    passenger_corrected_exposure: float

    heavy_corrected_exposure: float

    passenger_ordinary_values: np.ndarray

    heavy_ordinary_values: np.ndarray

    passenger_recurring_total: float

    heavy_recurring_total: float

    passenger_order_source: str

    heavy_order_source: str

    po_deterministic_summary: pd.DataFrame


def prepare_ordered_events(events: pd.DataFrame, case_label: str) -> tuple[pd.DataFrame,
str]:

    """

    Return included events for one case using a deterministic order.


    Best reproducibility practice:

    - If the mother workbook contains MC_Input_Order, Original_Input_Order, or
Simulation_Input_Order,

      the script uses that order.

    - If not, it uses Event_ID numerical order. This is deterministic, but it may not
reproduce an

      older Monte Carlo table if the original simulation used a different row order or
unrounded values.
```

```
    """

    required_cols = {"Case", "Event_ID", "Exposure_Category", "Value_USD",
"Included_In_Model"}

    missing = required_cols.difference(events.columns)

    if missing:

        raise KeyError(f"PO_Cleaned_Events is missing required columns:
{sorted(missing)}")



    included = events[true_like(events["Included_In_Model"])].copy()

    case_events = included[included["Case"].astype(str).str.strip().eq(case_label)].copy()



    order_candidates = ["MC_Input_Order", "Original_Input_Order",
"Simulation_Input_Order"]

    order_col = next((col for col in order_candidates if col in case_events.columns), None)



    if order_col is not None:

        case_events["_mc_order"] = pd.to_numeric(case_events[order_col], errors="coerce")

        if case_events["_mc_order"].isna().any():

            raise ValueError(f"{order_col} contains missing/non-numeric values for
{case_label}.")

        case_events = case_events.sort_values(["_mc_order", "Event_ID"],
kind="mergesort")

        order_source = f"{order_col} column"

    else:

        case_events["_event_id_number"] =
case_events["Event_ID"].map(event_id_number)

        case_events = case_events.sort_values(["_event_id_number", "Event_ID"],
kind="mergesort")
```

```
        order_source = "Event_ID numerical order (fallback; deterministic but may not match old MC order)"

    case_events["_value_usd"] = money_to_float(case_events["Value_USD"])
    return case_events.reset_index(drop=True), order_source


def calculate_po_results(workbook_path: Path) -> POResults:
    """Calculate purchase-order event values from PO_Cleaned_Events."""
    events = read_sheet_table(workbook_path, "PO_Cleaned_Events")

    passenger_events, passenger_order_source = prepare_ordered_events(events, PASSENGER)
    heavy_events, heavy_order_source = prepare_ordered_events(events, HEAVY)

    passenger_ordinary_mask = passenger_events["Exposure_Category"].astype(str).str.contains(
        "ordinary", case=False, na=False
    )
    heavy_ordinary_mask = heavy_events["Exposure_Category"].astype(str).str.contains(
        "ordinary", case=False, na=False
    )

    passenger_ordinary = passenger_events.loc[passenger_ordinary_mask].copy()
    passenger_recurring = passenger_events.loc[~passenger_ordinary_mask].copy()
    heavy_ordinary = heavy_events.loc[heavy_ordinary_mask].copy()
    heavy_recurring = heavy_events.loc[~heavy_ordinary_mask].copy()
```

```
    passenger_corrected = float(passenger_events["_value_usd"].sum())
    heavy_corrected = float(heavy_events["_value_usd"].sum())

    passenger_ordinary_values = passenger_ordinary["_value_usd"].to_numpy(dtype=float)
    heavy_ordinary_values = heavy_ordinary["_value_usd"].to_numpy(dtype=float)
    passenger_recurring_total = float(passenger_recurring["_value_usd"].sum())
    heavy_recurring_total = float(heavy_recurring["_value_usd"].sum())

    # Build deterministic PO summary directly from event table.
    rows = []
    for case_label, case_events, ordinary_events, recurring_events in [
        (PASSENGER, passenger_events, passenger_ordinary, passenger_recurring),
        (HEAVY, heavy_events, heavy_ordinary, heavy_recurring),
    ]:
        values = case_events["_value_usd"]
        total = float(values.sum())
        rows.append(
            {
                "case": case_label,
                "included_po_events": int(len(case_events)),
                "ordinary_latest_non_cancelled_po_events": int(len(ordinary_events)),
                "recurring_and_long_lead_events": int(len(recurring_events)),
                "ordinary_exposure_usd": float(ordinary_events["_value_usd"].sum()),
                "recurring_and_long_lead_exposure_usd":
float(recurring_events["_value_usd"].sum()),
                "corrected_po_exposure_usd": total,
```

```
            "mean_corrected_event_value_usd": float(values.mean()),
            "std_corrected_event_value_usd": float(values.std(ddof=1)),
            "coefficient_of_variation": float(values.std(ddof=1) / values.mean()),
            "top_five_concentration": float(values.nlargest(5).sum() / total),
        }
    )

    return POResults(
        passenger_corrected_exposure=passenger_corrected,
        heavy_corrected_exposure=heavy_corrected,
        passenger_ordinary_values=passenger_ordinary_values,
        heavy_ordinary_values=heavy_ordinary_values,
        passenger_recurring_total=passenger_recurring_total,
        heavy_recurring_total=heavy_recurring_total,
        passenger_order_source=passenger_order_source,
        heavy_order_source=heavy_order_source,
        po_deterministic_summary=pd.DataFrame(rows),
    )


def validate_deterministic_tables(workbook_path: Path, po_results: POResults) ->
pd.DataFrame:
    """Validate deterministic workbook values against the thesis values."""
    job_summary = read_summary_sheet(workbook_path, "Job_History", "Metric")
    spares_summary = read_summary_sheet(workbook_path, "SpareParts_Summary",
"Indicator")
    defects_summary = read_summary_sheet(workbook_path, "Defects_Summary",
"Defect indicator")
```

```
po_summary = read_summary_sheet(workbook_path, "PO_Summary", "Indicator")

rows = []

def add_check(name: str, actual: float, expected: float, tolerance: float = 0.01) -> None:
    difference = float(actual) - float(expected)
    status = "OK" if abs(difference) <= tolerance else "CHECK"
    rows.append(
        {
            "check": name,
            "actual": float(actual),
            "expected": float(expected),
            "difference": difference,
            "tolerance": tolerance,
            "status": status,
        }
    )
    assert_close(name, float(actual), float(expected), tolerance=tolerance)

# Job-history checks.
add_check(
    "Passenger job-history records",
    metric_value(job_summary, "Metric", "Job-history records", PASSENGER),
    EXPECTED_DETERMINISTIC["passenger_job_records"],
)
```

```
    add_check(
        "Heavy Lift job-history records",
        metric_value(job_summary, "Metric", "Job-history records", HEAVY),
        EXPECTED_DETERMINISTIC["heavy_job_records"],
    )
    add_check(
        "Passenger corrective job records",
        metric_value(job_summary, "Metric", "Corrective job records, based on Job Class",
PASSENGER),
        EXPECTED_DETERMINISTIC["passenger_corrective_jobs"],
    )
    add_check(
        "Heavy Lift corrective job records",
        metric_value(job_summary, "Metric", "Corrective job records, based on Job Class",
HEAVY),
        EXPECTED_DETERMINISTIC["heavy_corrective_jobs"],
    )
    add_check(
        "Passenger overdue job records",
        metric_value(job_summary, "Metric", "Overdue job records", PASSENGER),
        EXPECTED_DETERMINISTIC["passenger_overdue_jobs"],
    )
    add_check(
        "Heavy Lift overdue job records",
        metric_value(job_summary, "Metric", "Overdue job records", HEAVY),
        EXPECTED_DETERMINISTIC["heavy_overdue_jobs"],
```

```
    )

    # Purchase-order checks from summary sheet and from recalculated event table.
    add_check(
        "Passenger purchase-order rows",
        metric_value(po_summary, "Indicator", "Purchase-order rows", PASSENGER),
        EXPECTED_DETERMINISTIC["passenger_po_rows"],
    )
    add_check(
        "Heavy Lift purchase-order rows",
        metric_value(po_summary, "Indicator", "Purchase-order rows", HEAVY),
        EXPECTED_DETERMINISTIC["heavy_po_rows"],
    )
    add_check(
        "Passenger ordinary latest-version non-cancelled POs",
        metric_value(po_summary, "Indicator", "Ordinary latest-version non-cancelled POs",
PASSENGER),
        EXPECTED_DETERMINISTIC["passenger_ordinary_po_count"],
    )
    add_check(
        "Heavy Lift ordinary latest-version non-cancelled POs",
        metric_value(po_summary, "Indicator", "Ordinary latest-version non-cancelled POs",
HEAVY),
        EXPECTED_DETERMINISTIC["heavy_ordinary_po_count"],
    )
    add_check(
```

```
        "Passenger corrected PO exposure from event table",
        po_results.passenger_corrected_exposure,
        EXPECTED_DETERMINISTIC["passenger_corrected_exposure"],
    )
    add_check(
        "Heavy Lift corrected PO exposure from event table",
        po_results.heavy_corrected_exposure,
        EXPECTED_DETERMINISTIC["heavy_corrected_exposure"],
    )
    add_check(
        "Passenger recurring + long-lead exposure from event table",
        po_results.passenger_recurring_total,
        EXPECTED_DETERMINISTIC["passenger_recurring_exposure"] +
EXPECTED_DETERMINISTIC["passenger_long_lead_exposure"],
    )
    add_check(
        "Passenger ordinary PO exposure from event table",
        po_results.passenger_ordinary_values.sum(),
        EXPECTED_DETERMINISTIC["passenger_ordinary_exposure"],
    )
    add_check(
        "Heavy Lift ordinary PO exposure from event table",
        po_results.heavy_ordinary_values.sum(),
        EXPECTED_DETERMINISTIC["heavy_ordinary_exposure"],
    )
```

```
    # Spare-parts checks.
    add_check(
        "Passenger spare-parts records",
        metric_value(spares_summary, "Indicator", "Spare-parts records", PASSENGER),
        EXPECTED_DETERMINISTIC["passenger_spare_records"],
    )
    add_check(
        "Heavy Lift spare-parts records",
        metric_value(spares_summary, "Indicator", "Spare-parts records", HEAVY),
        EXPECTED_DETERMINISTIC["heavy_spare_records"],
    )
    add_check(
        "Passenger spare-parts records with stock above zero",
        metric_value(spares_summary, "Indicator", "Spare-parts records with stock above
zero", PASSENGER),
        EXPECTED_DETERMINISTIC["passenger_stock_above_zero"],
    )
    add_check(
        "Heavy Lift spare-parts records with stock above zero",
        metric_value(spares_summary, "Indicator", "Spare-parts records with stock above
zero", HEAVY),
        EXPECTED_DETERMINISTIC["heavy_stock_above_zero"],
    )
    add_check(
        "Passenger spare-parts stock-value indicator",
        metric_value(spares_summary, "Indicator", "Spare-parts stock-value indicator, USD",
PASSENGER),
```

```
        EXPECTED_DETERMINISTIC["passenger_stock_value"],
    )
    add_check(
        "Heavy Lift spare-parts stock-value indicator",
        metric_value(spares_summary, "Indicator", "Spare-parts stock-value indicator, USD", HEAVY),
        EXPECTED_DETERMINISTIC["heavy_stock_value"],
    )

    # Defect checks.
    add_check(
        "Heavy Lift defect records",
        metric_value(defects_summary, "Defect indicator", "Total defect records", HEAVY),
        EXPECTED_DETERMINISTIC["heavy_defect_records"],
    )

    return pd.DataFrame(rows)


# =====================================================================
# Monte Carlo simulation
# =====================================================================


def simulate_procurement_exposure(
    recurring_total: float,
    ordinary_values: np.ndarray,
    rng: np.random.Generator,
```

```
    runs: int = SIM_RUNS,
) -> np.ndarray:
    """
    Simulate procurement exposure.

    Recurring agreement exposure is retained as observed.
    Ordinary purchase-order exposure is simulated with:
    - Poisson event count, lambda = observed ordinary event count
    - sampling with replacement from observed ordinary purchase-order values
    """
    ordinary_values = np.asarray(ordinary_values, dtype=float)
    if ordinary_values.ndim != 1 or len(ordinary_values) == 0:
        raise ValueError("ordinary_values must be a non-empty one-dimensional array.")

    lam = len(ordinary_values)
    n_events = rng.poisson(lam=lam, size=runs)
    simulated_totals = np.empty(runs, dtype=float)

    for i, n in enumerate(n_events):
        if n > 0:
            simulated_totals[i] = recurring_total + rng.choice(
                ordinary_values,
                size=int(n),
                replace=True,
            ).sum()
```

```
        else:
            simulated_totals[i] = recurring_total

    return simulated_totals


def summarize_simulation(values: np.ndarray) -> dict[str, float]:
    """Return standard Monte Carlo summary statistics."""
    return {
        "mean": float(values.mean()),
        "median": float(np.median(values)),
        "standard_deviation": float(values.std(ddof=1)),
        "p5": float(np.percentile(values, 5)),
        "p95": float(np.percentile(values, 95)),
    }


def run_monte_carlo(po_results: POResults) -> tuple[pd.DataFrame, pd.DataFrame,
pd.DataFrame]:
    """
    Run Monte Carlo simulation using independent RNG streams.

    Independent streams make the Passenger Vessel result independent of whether the
Heavy Lift
    simulation is run before/after it. This improves reproducibility and auditability.
    """
    seed_sequence = np.random.SeedSequence(RANDOM_SEED)
    passenger_seed, heavy_seed = seed_sequence.spawn(2)
```

```
passenger_rng = np.random.default_rng(passenger_seed)
heavy_rng = np.random.default_rng(heavy_seed)

passenger_values = simulate_procurement_exposure(
    recurring_total=po_results.passenger_recurring_total,
    ordinary_values=po_results.passenger_ordinary_values,
    rng=passenger_rng,
    runs=SIM_RUNS,
)
heavy_values = simulate_procurement_exposure(
    recurring_total=po_results.heavy_recurring_total,
    ordinary_values=po_results.heavy_ordinary_values,
    rng=heavy_rng,
    runs=SIM_RUNS,
)
difference_values = passenger_values - heavy_values

passenger_summary = summarize_simulation(passenger_values)
heavy_summary = summarize_simulation(heavy_values)
difference_summary = summarize_simulation(difference_values)

mc_summary = pd.DataFrame(
    [
        {"case": PASSENGER, **passenger_summary},
        {"case": HEAVY, **heavy_summary},
```

```
            {"case": "Difference, Passenger minus Heavy Lift", **difference_summary},
        ]
    )

    probability_summary = pd.DataFrame(
        [
            {
                "probability_passenger_exposure_higher": float((passenger_values >
heavy_values).mean()),
                "probability_heavy_lift_exposure_higher": float((heavy_values >
passenger_values).mean()),
            }
        ]
    )

    samples = pd.DataFrame(
        {
            "run_id": np.arange(1, SIM_RUNS + 1),
            "passenger_exposure_usd": passenger_values,
            "heavy_lift_exposure_usd": heavy_values,
            "difference_passenger_minus_heavy_usd": difference_values,
        }
    )

    return mc_summary, probability_summary, samples
```

```
def compare_with_reported_mc(mc_summary: pd.DataFrame) -> pd.DataFrame:
    """Compare recalculated MC values with the older reported thesis MC values."""
    passenger = mc_summary.loc[mc_summary["case"].eq(PASSENGER)].iloc[0]
    heavy = mc_summary.loc[mc_summary["case"].eq(HEAVY)].iloc[0]
    diff = mc_summary.loc[mc_summary["case"].str.startswith("Difference")].iloc[0]

    comparisons = [
        ("Passenger mean", passenger["mean"], REPORTED_MC["passenger_mean"]),
        ("Passenger median", passenger["median"], REPORTED_MC["passenger_median"]),
        ("Passenger standard deviation", passenger["standard_deviation"],
REPORTED_MC["passenger_std"]),
        ("Passenger p5", passenger["p5"], REPORTED_MC["passenger_p5"]),
        ("Passenger p95", passenger["p95"], REPORTED_MC["passenger_p95"]),
        ("Heavy Lift mean", heavy["mean"], REPORTED_MC["heavy_mean"]),
        ("Heavy Lift median", heavy["median"], REPORTED_MC["heavy_median"]),
        ("Heavy Lift standard deviation", heavy["standard_deviation"],
REPORTED_MC["heavy_std"]),
        ("Heavy Lift p5", heavy["p5"], REPORTED_MC["heavy_p5"]),
        ("Heavy Lift p95", heavy["p95"], REPORTED_MC["heavy_p95"]),
        ("Difference mean", diff["mean"], REPORTED_MC["diff_mean"]),
        ("Difference p5", diff["p5"], REPORTED_MC["diff_p5"]),
        ("Difference p95", diff["p95"], REPORTED_MC["diff_p95"]),
    ]

    rows = []
    for metric, recalculated, reported in comparisons:
```

```
        difference = float(recalculated) - float(reported)
        rows.append(
            {
                "metric": metric,
                "recalculated_from_mother_doc": float(recalculated),
                "reported_old_thesis_value": float(reported),
                "difference": difference,
                "absolute_difference": abs(difference),
            }
        )

    comparison_df = pd.DataFrame(rows)

    if STRICT_REPORTED_MC_VALIDATION:
        for _, row in comparison_df.iterrows():
            assert_close(
                row["metric"],
                row["recalculated_from_mother_doc"],
                row["reported_old_thesis_value"],
                tolerance=0.01,
            )

    return comparison_df

# ==========================================================
```

```
# Sensitivity calculations
# ============================================================

def calculate_sensitivity(po_results: POResults, workbook_path: Path) -> pd.DataFrame:
    """Calculate break-even and sensitivity checks."""
    spares_summary = read_summary_sheet(workbook_path, "SpareParts_Summary", "Indicator")
    passenger_stock_value = float(
        metric_value(spares_summary, "Indicator", "Spare-parts stock-value indicator, USD", PASSENGER)
    )
    heavy_stock_value = float(
        metric_value(spares_summary, "Indicator", "Spare-parts stock-value indicator, USD", HEAVY)
    )

    passenger_po = po_results.passenger_corrected_exposure
    heavy_po = po_results.heavy_corrected_exposure

    heavy_break_even_factor = passenger_po / heavy_po
    passenger_reduction_factor = heavy_po / passenger_po
    spare_parts_weight_factor = (passenger_po - heavy_po) / (heavy_stock_value - passenger_stock_value)

    return pd.DataFrame(
        [
            {
```

```
            "sensitivity_test": "Heavy Lift Vessel purchase-order break-even factor",
            "result": heavy_break_even_factor,
            "rounded_thesis_result": round(heavy_break_even_factor, 2),
            "interpretation": "Heavy Lift exposure factor needed to equal Passenger corrected PO exposure.",
        },
        {
            "sensitivity_test": "Passenger Vessel purchase-order reduction factor",
            "result": passenger_reduction_factor,
            "rounded_thesis_result": round(passenger_reduction_factor, 2),
            "interpretation": "Passenger exposure share needed to equal Heavy Lift corrected PO exposure.",
        },
        {
            "sensitivity_test": "Spare-parts stock-value break-even factor",
            "result": spare_parts_weight_factor,
            "rounded_thesis_result": round(spare_parts_weight_factor, 2),
            "interpretation": "Spare-parts stock exposure weight needed before overturning PO exposure difference.",
        },
        {
            "sensitivity_test": "Recorded downtime sensitivity",
            "result": np.nan,
            "rounded_thesis_result": np.nan,
            "interpretation": "Inconclusive because both cases have zero positive recorded downtime hours.",
        },
```

```
        ]
    )


# ============================================================
# Main workflow
# ============================================================


def write_audit_note(
    workbook_path: Path,
    po_results: POResults,
    mc_comparison: pd.DataFrame,
) -> None:
    """Write a plain-text audit note explaining exact reproducibility status."""
    OUTPUT_DIR.mkdir(exist_ok=True)

    max_abs_mc_difference = float(mc_comparison["absolute_difference"].max())

    note = f"""Reproducibility audit note
==========================


Input workbook: {workbook_path}
Input workbook SHA-256: {file_hash(workbook_path)}
Python version: {platform.python_version()}
pandas version: {pd.__version__}
NumPy version: {np.__version__}
```

Random seed: {RANDOM_SEED}

Simulation runs: {SIM_RUNS}

Monte Carlo input order

Passenger Vessel order source: {po_results.passenger_order_source}

Heavy Lift Vessel order source: {po_results.heavy_order_source}

Important note

The deterministic workbook calculations are validated against the thesis values.

The Monte Carlo simulation is now reproducible because it uses deterministic event ordering

and independent random-number streams.

If the recalculated Monte Carlo table differs from an older thesis table, that does not mean

the deterministic calculations are wrong. It means the old Monte Carlo output was generated

from a specific input order and/or full-precision values that may not be fully preserved in the

anonymized mother workbook. To reproduce the old Monte Carlo output exactly, add an

MC_Input_Order column and use full-precision Value_USD values from the original model input.

Maximum absolute difference versus older reported MC table: {max_abs_mc_difference:.6f} USD

"""

```
    (OUTPUT_DIR / "reproducibility_audit_note.txt").write_text(note, encoding="utf-8")


def main() -> None:
    workbook_path = locate_workbook()
    OUTPUT_DIR.mkdir(exist_ok=True)

    print("Environment")
    print("-----------")
    print(f"Python version: {platform.python_version()}")
    print(f"pandas version: {pd.__version__}")
    print(f"NumPy version: {np.__version__}")
    print(f"Random seed: {RANDOM_SEED}")
    print(f"Simulation runs: {SIM_RUNS}")
    print()

    print("Input-file integrity")
    print("--------------------")
    print(f"Workbook: {workbook_path}")
    print(f"SHA-256: {file_hash(workbook_path)}")
    print()

    po_results = calculate_po_results(workbook_path)
    deterministic_checks = validate_deterministic_tables(workbook_path, po_results)
    sensitivity = calculate_sensitivity(po_results, workbook_path)
    mc_summary, probability_summary, mc_samples = run_monte_carlo(po_results)
```

```
    mc_comparison = compare_with_reported_mc(mc_summary)


    # Export outputs.

    deterministic_checks.to_csv(OUTPUT_DIR / "deterministic_validation_checks.csv",
index=False)

    po_results.po_deterministic_summary.to_csv(OUTPUT_DIR /
"po_deterministic_summary.csv", index=False)

    sensitivity.to_csv(OUTPUT_DIR / "sensitivity_checks.csv", index=False)

    mc_summary.to_csv(OUTPUT_DIR / "monte_carlo_summary_recalculated.csv",
index=False)

    probability_summary.to_csv(OUTPUT_DIR / "monte_carlo_probability_summary.csv",
index=False)

    mc_comparison.to_csv(OUTPUT_DIR /
"monte_carlo_comparison_vs_old_reported.csv", index=False)

    mc_samples.to_csv(OUTPUT_DIR / "monte_carlo_samples_recalculated.csv",
index=False)

    write_audit_note(workbook_path, po_results, mc_comparison)


    # Console output.

    print("All deterministic validation checks passed.")

    print()

    print("PO deterministic summary")

    print("------------------------")

    print(po_results.po_deterministic_summary.to_string(index=False))

    print()

    print("Sensitivity checks")

    print("------------------")

    print(sensitivity.to_string(index=False))
```

```
    print()
    print("Monte Carlo summary, recalculated")
    print("--------------------------------")
    print(mc_summary.to_string(index=False))
    print()
    print("Monte Carlo probability summary")
    print("------------------------------")
    print(probability_summary.to_string(index=False))
    print()

    max_abs_diff = float(mc_comparison["absolute_difference"].max())
    if max_abs_diff > 0.01:
        warnings.warn(
            "The recalculated Monte Carlo output differs from the older reported MC table. "
            "This is expected if the anonymized workbook lacks original MC_Input_Order or full-precision values. "
            "See reproducibility_output/reproducibility_audit_note.txt.",
            RuntimeWarning,
        )
        print("Monte Carlo comparison versus old reported table")
        print("----------------------------------------------")
        print(mc_comparison.to_string(index=False))
        print()

    print(f"Output files saved to: {OUTPUT_DIR.resolve()}")
```

```python
if __name__ == "__main__":
    main()
```